

# Lightweight vision-based 6D pose estimation of curtain wall module installation slot

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## ABSTRACT

Accurately localizing and estimating the 6D pose of the target installation slot is a core perception requirement for robot-assisted Curtain Wall Module (CWM) installation in high-rise construction. Current practice often relies on visual judgment and verbal coordination during alignment at height, which is time-consuming, error-prone, and safety-critical. This paper presents a lightweight, markerless, monocular vision pipeline for target-slot localization and 6D pose estimation. The proposed two-stage pipeline integrates a lightweight facade semantic segmentation network with geometric corner extraction and a Perspective-n-Point (PnP) solver. First, the segmentation model parses the facade into structural components, providing stable geometric cues. Second, a rule-based corner-extraction and ordering procedure constructs 2D-3D correspondences, which are fused with a predefined 3D CWM model via PnP to estimate the 6D pose of an installed reference panel. The target-slot pose is then obtained through deterministic pose propagation using the facade layout and known module spacing. In the MuJoCo simulation, across 36 target-slot cases, the proposed method achieves a median 3D translation error of 11.59 mm (mean 16.34 mm), a median planar error of 2.24 mm (mean 3.08 mm), and a median orientation error of 4.91° (mean 7.19°). Compared with an AprilTag fiducial baseline, the method attains comparable in-plane target-slot localization and competitive orientation estimates while avoiding marker dependency.

## 1. Introduction

With advances in robotics and sensing, construction automation is increasingly shifting from manual, labor-intensive operations toward machine-assisted workflows with higher autonomy (Chen et al., 2018b; Wu et al., 2022). In this transition, machine vision plays a central role by enabling robots to perceive the built environment and localize task-relevant targets, making automated assembly and installation a practical and scalable direction for on-site deployment (Halder & Afsari, 2023; Ouda & Haggag, 2024). Accordingly, robotic installation and assembly systems can substantially improve productivity and safety relative to traditional manual methods, particularly for hazardous facade operations performed at height (Behnam et al., 2023). Nevertheless, despite its central role in the global economy, the construction sector remains comparatively under-automated, with persistent productivity challenges and limited diffusion of advanced manufacturing-style automation on job sites. The "Construction4.0" agenda argues that digitization, sensing, and cyber-physical systems can bridge this gap by enabling data-driven planning and automated execution (Oesterreich & Teuteberg, 2016). In parallel, recent reviews show accelerating uptake of artificial intelli-

gence (AI) across construction tasks—from vision-based monitoring and quality control to decision support—yet also highlight that real-world deployment requires robust perception, reliability, and integration with field workflows (Abioye et al., 2021; Wang et al., 2026).

Modular construction, especially unitized CWM installation in high-rise buildings, exemplifies the need for practical automation. Conventional installation involves transporting and aligning heavy modules at height, typically relying on crane operators and crews to coordinate installation through visual judgment and verbal communication. This process is labor-intensive and time-critical, and it remains safety-sensitive under wind loads, glare, occlusions, and restricted access. Automating or robotically assisting CWM installation can reduce worker exposure in hazardous zones, shorten alignment cycles, and improve placement consistency, thereby enhancing both productivity and site safety. Prior work has therefore identified unitized CWM installation as a high-impact target for automation and reviewed existing solutions and open challenges for unitized CWM installation (Johns et al., 2020).

Recent work on robotized automated installation of CWM concepts has demonstrated the feasibility of mechanized handling and assisted placement of CWMs (Iturralde et al., 2022; Yu et al., 2007).

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For example, a cable-driven parallel robot has been developed and evaluated for CWM installation, demonstrating both the potential for improved productivity and the practical constraints arising from accuracy requirements and system integration (Iturralde et al., 2022). More broadly, construction robotics research emphasizes that progress depends not only on hardware capability, but also on robust autonomous perception-planning-control stacks and credible pathways for real-world adoption. However, current automated CWM installation systems still lack such perception capability to reliably localize the target slot (Hassen et al., 2025) and estimate its 6D pose (Davila Delgado et al., 2019). Nevertheless, reliable localization and pose estimation for CWM installation remain challenging (Wu et al., 2024). Facade scenes often contain repetitive panel patterns, strong specular reflections from glass, partial occlusions caused by lifting gear or adjacent installed modules, and substantial illumination variations across time of day and viewing angles. These factors can undermine purely geometric methods (e.g., edge- or corner-based detection). At the same time, many end-to-end pose pipelines are computationally intensive and data-hungry, which complicates real-time deployment on construction hardware and increases the annotation burden for pose-supervised learning.

Another widely used vision-based approach is marker-based pose estimation, which infers 6D pose by detecting artificial fiducials (e.g., AprilTag/ArUco) and solving a geometric pose-recovery problem from the known marker geometry (Kalaitzakis et al., 2021). Marker-based pose estimation is often impractical for curtain wall installation because it requires extensive fiducial placement and calibration. Recent markerless single-RGB correspondence methods can provide more practical and robust CWM pose estimation by deriving 2D-3D correspondence from learned features and geometry without site instrumentation.

Motivated by these challenges, this paper proposes a segmentation-based, markerless pose estimation system for robot-assisted CWM installation that combines a lightweight segmentation model with geometric reasoning to estimate the 6D pose of the target installation slot. Unlike end-to-end pose regression, the proposed pipeline explicitly exploits the geometric structure of unitized CWMs, such as known module dimensions and a regular facade layout, to infer the target-slot pose from the segmentation output. To benchmark performance, we compare the proposed markerless estimator against a marker-based AprilTag reference baseline. Results show competitive translation and orientation accuracy across most test cases, while eliminating fiducial deployment and associated site instrumentation.

The key contributions of this paper are summarized as follows:

- 1) A robot-oriented, lightweight markerless monocular perception module for curtain-wall installation that localizes the target installation slot and outputs a 6D pose in a control-ready coordinate frame for downstream alignment and motion planning.
- 2) A segmentation-driven CWM instance corner extraction and refinement to recover stable corner correspondences in cluttered facade imagery.
- 3) A single-RGB markerless 6D pose estimation pipeline tailored to curtain-wall installation, coupling the ordered corners with a planar PnP solver and explicit validity checks to accept/reject pose estimates.
- 4) A quantitative evaluation against an AprilTag fiducial reference baseline, reporting translation and orientation errors across multiple target-slot cases and viewpoint variations.

The remainder of this paper is organised as follows: Section 2 reviews related work; Section 3 presents the proposed approach; Section 4 details the experimental setup and evaluation metrics; Section 5 reports results and discussion; and Section 6 concludes with limitations and future work.

## 2. Related work

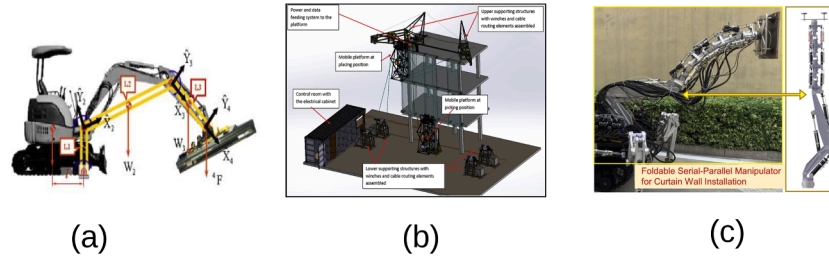
### 2.1. Automated curtain wall installation

The installation of CWM remains largely manual, hazardous, and time-consuming. However, as automated systems become more capable and reliable, automation in construction is expanding, thereby reducing manual labor and risk exposure in high-risk environments. Recent research on curtain wall systems spans the full life-cycle, encompassing facade and curtain wall design and performance assessment (Formisano & Naqash, 2026; Luo et al., 2025), robot-assisted CWMs installation (Iturralde et al., 2022; Johns et al., 2023a; Taghavi et al., 2018; Wu et al., 2024), and automated maintenance applications such as curtain wall cleaning (Lee et al., 2018) and facade inspection robots (Halder & Afshari, 2023). Despite these advances, on-site CWM installation remains challenging due to unstructured site conditions, crane-worker coordination, and tight geometric tolerances, which often necessitate multiple workers to manually perform handling, positioning, and alignment tasks (Taghavi et al., 2018). Consequently, a growing body of work has pursued robotic and digital solutions, accelerating the shift from traditional manual methods toward automated and semi-automated installation processes. Robot-assisted installation systems for CWMs have demonstrated the potential to support lifting, handling, and fine alignment, improving safety and reducing physical workload during high-rise CWM installation; representative automated robotic systems in this area are shown in Fig. 1. Nevertheless, many of these platforms still lack an integrated, automated capability to localize and estimate the 6D pose of the target installation slot, which remains a key prerequisite for a truly autonomous CWM installation system.

### 2.2. Lightweight semantic segmentation

Semantic segmentation is a fundamental computer vision task with wide-ranging real-world applications, including autonomous driving, video surveillance, medical image analysis, robotics, and augmented reality. Its prominence has grown substantially with advances in deep learning-based methods (Cărunta et al., 2025; Liang et al., 2026; Wang et al., 2026). Deep learning-based semantic segmentation methods can be classified into two broad categories: *heavyweight* architectures that prioritize peak accuracy, and *lightweight* architectures designed for efficient, on-edge-device deployment. A key drawback of many heavyweight deep learning architectures (Chen et al., 2015, 2018a; Guo et al., 2022; Liu et al., 2022; Ronneberger et al., 2015; Wang et al., 2020; Zheng et al., 2021) is their limited practicality for deployment on edge devices. Although they deliver impressive accuracy, their high computational cost often leads to long inference times and substantial hardware requirements. This motivates the development of lightweight models (Hong et al., 2021; Oršić & Šegvić, 2021; Paszke et al., 2016; Peng et al., 2022; Shi et al., 2022; Yu et al., 2021) that preserve competitive segmentation performance while enabling low-latency, high-throughput inference on resource-constrained edge devices. These lightweight designs have also been validated for defect/crack segmentation (e.g., tunnel crack detection and real-time surface defect segmentation) (Chen et al., 2023; Huang et al., 2026).

Lightweight semantic segmentation research primarily targets real-time, resource-constrained deployment (e.g., edge CPUs/GPUs) by reducing parameter count, memory footprint, and latency through architecture-level efficiency. In this regard, two architecture design paradigms dominate: (i) single-stream networks (Ronneberger et al., 2015; Yu et al., 2018), which balance semantics and spatial detail; and (ii) multi-stream networks (Hong et al., 2021; Xu et al., 2023; Yu et al., 2018), which process spatial detail and semantic context in parallel to improve accuracy-speed trade-offs. In response to this trade-off, recent work increasingly favors single-stream lightweight backbones that consolidate feature extraction and context modeling into a single pipeline. In this context, LAnet (Hassen et al., 2025) is a strong representative of



**Fig. 1.** Developed CWM installation robots. (a) Curtain wall installation robot (Yu et al., 2007). (b) Cable-driven parallel robot for CWMs automatic installation (Iturralde et al., 2022). (c) A glass curtain wall installation robotic arm (Liu et al., 2025).

lightweight semantic segmentation networks for CWM installation, as it is trained and tested for curtain wall installation scenarios.

### 2.3. Pose estimation

For autonomous or machine-assisted curtain wall installation, accurate target slot localization and 6D pose estimation of the intended installation slot are essential to enable reliable motion planning, alignment, and closed-loop placement control (Chen et al., 2026; Zhang et al., 2026). Unlike factory pick-and-place systems that operate in structured, repeatable environments, construction robots must perform pose estimation under uncontrolled site conditions and visually challenging environments. Consequently, mature methods developed and validated in highly controlled industrial settings may not transfer reliably to construction environments, as their assumptions are frequently challenged in construction environments. In construction, pose estimation methods are commonly categorized into sensor-based and vision-based. Sensor-based pose estimation often leverages LiDAR-IMU to produce drift-controlled 6D trajectories in cluttered, GNSS-denied, and dynamically changing sites (Lee, 2024). At the same time, IMU-only kinematic pipelines have also been demonstrated for full-body pose tracking of articulated construction machinery (Feng et al., 2025; Stührenberg & Smarsly, 2025; Tang et al., 2022). Vision-based approaches estimate 3D pose directly from monocular/RGB-D imagery (Boldo et al., 2024) and increasingly employ construction-tailored visual SLAM that integrates dynamic-object segmentation and scene semantics to maintain robust tracking in non-stationary workspaces (Kim et al., 2023; Yang & Cai, 2024).

Another widely used vision method is *marker-based* pose estimation, which infers 6D pose by detecting artificial fiducials (e.g., April-Tag/ArUco) and solving a geometric pose recovery problem from the known marker geometry (Deng et al., 2025; Kalaitzakis et al., 2021; Kayhani et al., 2022). However, despite their effectiveness in many domains, marker-based vision methods are often impractical for curtain wall installation robots because they require attaching and maintaining fiducials on CWMs across large work areas and introduce additional site preparation and calibration overhead. Likewise, active range sensors such as LiDAR can be unreliable on glass and other specular facade materials due to multi-path reflections and refractive effects, which may yield invalid returns or depth readings that correspond to reflected/background surfaces rather than the true installation interface.

Recent markerless keypoint correspondence-based vision methods using a single RGB image offer a more practical and potentially more robust alternative for target installation slot pose estimation, as they can exploit semantic information and geometric constraints to establish 2D-3D correspondences with already available information. Correspondence-based 6D object pose estimation seeks to recover an object's pose by establishing correspondences between observed input data and a known 3D object model. A closely related study by Wu et al. (2024) proposes a dual-flow aggregation network (DANF) for curtain-wall frame segmentation and further designs a pose-solving method based on the segmented frame class to support curtain-wall installation

scenarios. Unlike DANF's frame-centric pose recovery, our work targets the 6D pose of the target installation slot on a building facade, using a pre-installed module as a reference.

## 3. Methodology

### 3.1. Overview

Algorithmically, the proposed pipeline is hybrid and modular: (i) a lightweight semantic segmentation network provides a structured scene understanding of glass, frame, and background regions, and (ii) a rule-based method extracts stable ordered 2D corner correspondences from segmentation output, which are then lifted to a 6D pose via a PnP solver. Accordingly, our approach closely aligns with keypoint-based monocular pose estimation since we recover pose via PnP from ordered 2D corners. This design fundamentally differs from end-to-end monocular pose networks, which learn a direct RGB-to-pose mapping, typically require large pose-labeled datasets, and can be sensitive to construction-specific application shifts.

Given an input RGB image of a building facade, our goal is to detect installed CWMs and estimate the 6D pose of the target installation slot, represented by three translational and three rotational components. We assume each CWM is rigid and its physical dimensions are known *a priori*. As illustrated in Fig. 2, first, a semantic segmentation network predicts three classes (*glass*, *frame*, *background*). Then, the predicted masks are post-processed using a rule-based geometric pipeline to separate adjacent panel instances, extract consistently ordered 2D corner keypoints, and solve a planar PnP problem with a known panel model to recover the 6D pose of a reference panel. The target installation-slot pose is then inferred by deterministic geometric translation along the local panel-width direction in the camera frame. As shown in Fig. 3, the overall pipeline includes failure detection and recovery mechanisms to handle pose-related failures. The main methodological contributions of this paper lie in the segmentation-to-pose pipeline built on top of the segmentation output. In the following subsections, we briefly summarize LANet and then describe our proposed rule-based post-processing for 2D keypoint recovery and the PnP-based pose estimation procedure.

### 3.2. LANet

In this paper, we adopt LANet (Fig. 4), a single-stream semantic segmentation network designed to segment building facades into three classes: *frame*, *glass*, and *background*. LANet is designed to achieve a favorable accuracy-efficiency trade-off by avoiding heavy encoder-decoder designs and redundant multi-branch architectures, while retaining sufficient contextual reasoning for facade parsing. The network is compact (1.92M parameters) and achieves real-time inference performance (262 FPS on an RTX 3090 GPU and 19 FPS on an Intel i7 CPU). For full architectural details, ablation studies, and comparisons with the state-of-the-art lightweight models, we refer the reader to the LANet paper (Hassen et al., 2025).

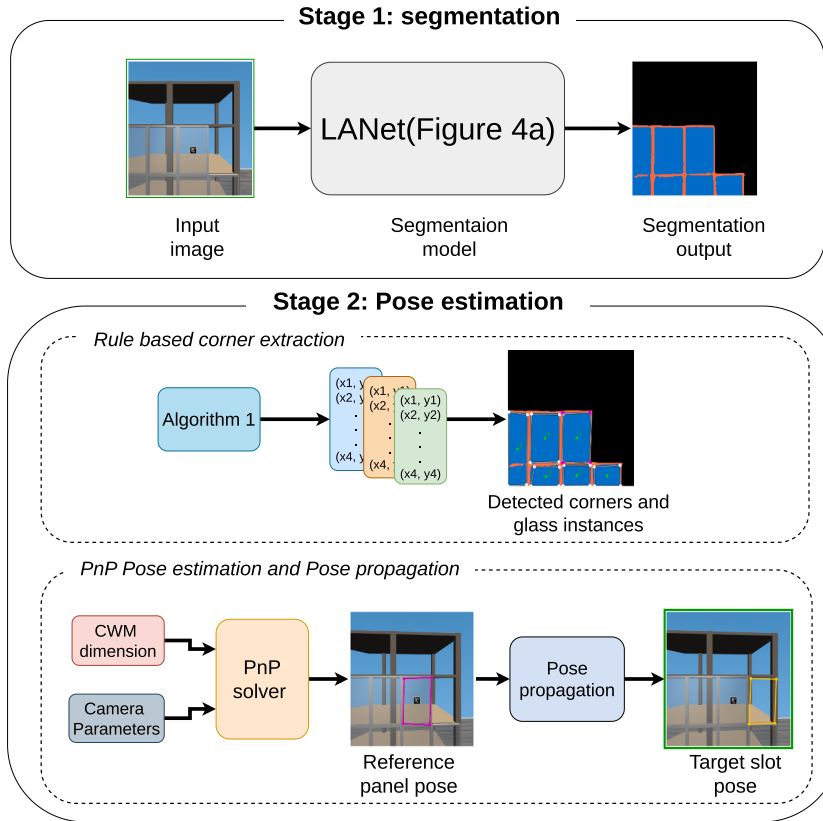


Fig. 2. Overall pipeline of our methodology.

### 3.3. Rule-based corner detection

The proposed corner-extraction stage is designed for rectangular curtain-wall modules. Specifically, the algorithm enforces a quadrilateral hypothesis and outputs four ordered corners to construct 2D–3D correspondences for a planar PnP solver under known panel geometry. Consequently, the current implementation is not directly applicable to non-rectangular panels without redefining the correspondence model. In this work, we recover reference-panel corners from the segmentation output by using the predicted frame as a structural barrier to separate adjacent glass regions, followed by robust quadrilateral fitting and consistent corner ordering for each extracted glass instance. In [Algorithm 1](#),  $S \in \{0, 1, 2\}^{H \times W}$  denotes the predicted semantic label map (glass/frame/background) with image size  $H \times W$ . We form binary masks  $M_g = \mathbb{1}[S = c_g]$  and  $M_f = \mathbb{1}[S = c_f]$ , where  $c_g$  and  $c_f$  are the class IDs for glass and frame classes, respectively, and  $\mathbb{1}[\cdot]$  is the indicator function. The refined frame barrier is denoted by  $B_f \in \{0, 1\}^{H \times W}$  and is obtained by applying directional morphological closing with kernel length  $k_{\text{close}}$ , followed by opening, and then dilation with kernel size  $k_{\text{dilate}}$ . Removing the barrier from the glass mask yields a split glass mask  $M_g^{\text{split}} = M_g \odot (1 - B_f)$ , where  $\odot$  is element-wise multiplication. Connected-component labeling on  $M_g^{\text{split}}$  produces an instance map  $I \in \{0, 1, \dots, N\}^{H \times W}$ , where  $N$  is the number of detected glass instances and  $I(\mathbf{p}) = \ell$  assigns pixel  $\mathbf{p}$  to instance  $\ell$ . For each instance  $\ell$ , we define its binary mask  $M_\ell = \mathbb{1}[I = \ell]$  and dilate it with a marker kernel of size  $k_M$  to obtain a reclaim region  $\tilde{M}_\ell$  used to recover boundary glass pixels from  $M_g$  while excluding frame pixels. Contours are extracted from each reclaimed instance mask using OpenCV. A minimum-area rectangle is fitted to obtain a quadrilateral  $\mathbf{Q}_\ell = \{\mathbf{q}_{\ell,i}\}_{i=1}^4$  with vertices ordered as top-left (TL)–top-right (TR)–bottom-right (BR)–bottom-left (BL). We discard instances whose pixel area is below  $\tau_{\text{min}}$ . The reference panel is selected as the instance with the smallest centroid  $y$  (top-most) and the

largest centroid  $x$  (right-most). Its ordered corners  $\mathbf{Q}_{\text{ref}}$  are used as the 2D corner correspondences as input to the PnP algorithm.

### 3.4. Pose estimation using PnP algorithm

After refining the reference-panel corners using the rule-based corner detection in [Section 3.3](#), we estimate the reference panel's 6D pose relative to the camera frame by solving a PnP problem. Let  $\mathbf{u}_i = [u_i, v_i]^T$ ,  $i \in \{1, \dots, 4\}$ , corresponds to the glass class corner keypoints in {TL, TR, BR, BL} order. Because the CWM geometry is known, we define the associated 3D corner coordinates in the object frame as a planar rectangle:

$$\mathbf{X}_i = \left[ \pm \frac{w_g}{2}, \pm \frac{h_g}{2}, 0 \right]^T, \quad i \in \{1, \dots, 4\}, \quad (1)$$

where  $w_g$  and  $h_g$  denote the glass width and height.

Given the camera intrinsic matrix  $\mathbf{K}$ , PnP estimates the rotation  $\mathbf{R}_{co} \in SO(3)$  and translation  $\mathbf{t}_{co} \in \mathbb{R}^3$  that map points from the object frame to the camera frame by minimizing the reprojection error

$$\min_{\mathbf{R}_{co}, \mathbf{t}_{co}} \sum_{i=1}^4 \left\| \pi(\mathbf{K}(\mathbf{R}_{co}\mathbf{X}_i + \mathbf{t}_{co})) - \mathbf{u}_i \right\|_2^2, \quad (2)$$

where  $\pi(\cdot)$  denotes the perspective division that converts homogeneous coordinates to image coordinates. In our implementation, we estimate pose from the four coplanar corners using OpenCV PnP. The resulting pose  $\{\mathbf{R}_{co}, \mathbf{t}_{co}\}$  provides the reference panel pose in the camera coordinate frame. The target installation slot is inferred by translating the reference pose along the local width axis expressed in the camera frame. Specifically, letting  $\hat{\mathbf{x}}_c$  denote the first column of  $\mathbf{R}_{co}$  (the local glass-width direction in the camera frame), the next glass center is predicted as

$$\mathbf{t}_c^{\text{next}} = \mathbf{t}_{co} + (w_g + g) \hat{\mathbf{x}}_c, \quad (3)$$



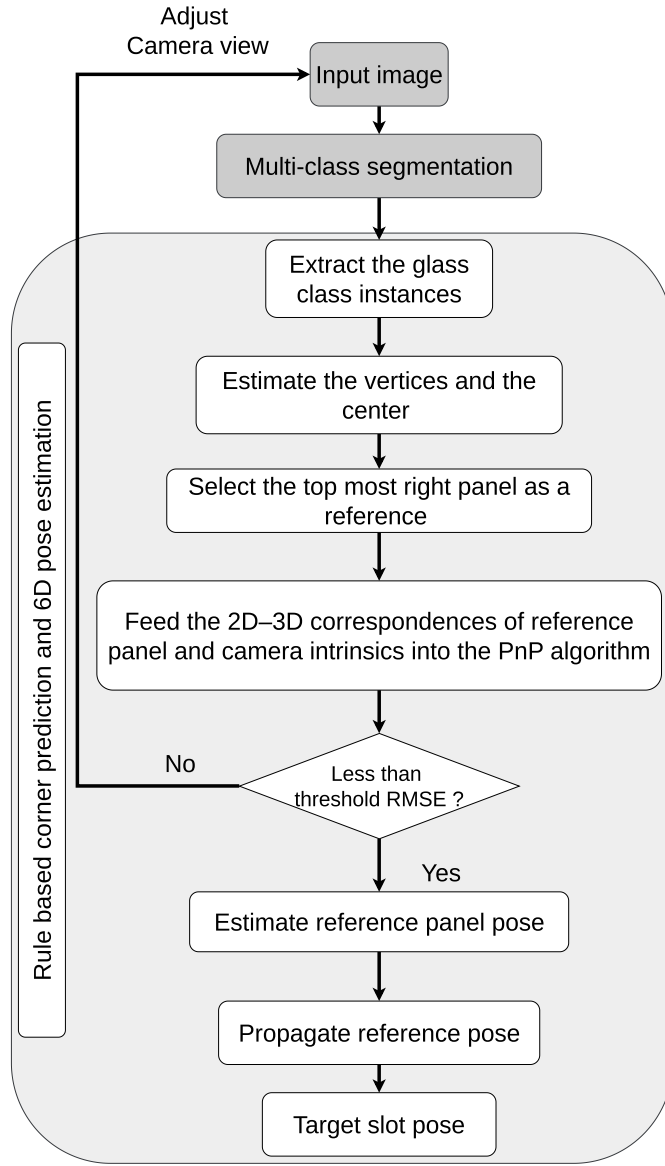


Fig. 3. Overview of the visual pose estimation pipeline.

where  $g$  is the inter-panel *glass* gap. For simplicity, the next pose orientation is kept the same as the reference orientation, i.e.,  $\mathbf{R}_c^{\text{next}} = \mathbf{R}_{co}$ .

### 3.5. Dataset preparation

We curated a custom dataset comprising real facade imagery and physics-based synthetic images to support both segmentation training and downstream 6D pose estimation. Real images were collected from two primary sources: (i) publicly available images through web scraping and (ii) images captured using a lab-scale building prototype (Johns et al., 2023b). The web images predominantly depict completed buildings and provide substantial visual diversity, including a wide range of architectural styles, structural layouts, and lighting conditions (Fig. 5a). To ensure task relevance and to capture robotic installation scenarios, additional images were collected using a physical lab-scale building prototype (Fig. 5b). All images were manually annotated at the pixel level into three semantic classes: *Frame* (structural elements such as curtain wall frames, mullions, and transoms), *Glass* (transparent or translucent infill regions), and *Background*. The resulting real dataset

**Algorithm 1** Segmentation-driven instance extraction and reference panel corner estimation.

**Require:** Segmentation label map  $M \in \mathbb{R}^{H \times W}$ , class IDs ( $c_g, c_f$ ) for glass/frame  
**Ensure:** Candidate panel corners  $Q = \{Q_i\}$  where  $Q_i \in \mathbb{R}^{4 \times 2}$  (ordered TL, TR, BR, BL), and reference panel corners  $Q_{\text{ref}}$

- 1:  $G \leftarrow \mathbb{1}[M = c_g]$ ,  $F \leftarrow \mathbb{1}[M = c_f]$   $\triangleright$  binary glass and frame masks
- 2:  $\tilde{F} \leftarrow \text{DIRECTIONALCLOSE}(F, k_{\text{close}})$   $\triangleright$  bridge gaps in mullions/transoms
- 3:  $\tilde{F} \leftarrow \text{OPEN}(\tilde{F}, 3)$   $\triangleright$  remove small noise
- 4:  $B \leftarrow \text{DILATE}(\tilde{F}, k_{\text{dilate}})$   $\triangleright$  refined frame mask,  $B \in \{0, 1\}^{H \times W}$
- 5:  $G_s \leftarrow G \wedge \neg B$   $\triangleright$  split-glass mask to prevent merged panels
- 6:  $L \leftarrow \text{CONNECTEDCOMPONENTS}(G_s)$   $\triangleright$  instance labels  $1, \dots, N$
- 7:  $Q \leftarrow \emptyset$
- 8: **for**  $i = 1$  to  $N$  **do**
- 9:  $S_i \leftarrow \mathbb{1}[L = i]$
- 10: **if**  $|S_i| < \tau_{\text{min}}$  **then**
- 11: **continue**
- 12: **end if**
- 13:  $S_i^+ \leftarrow \text{DILATE}(S_i, k_M)$   $\triangleright$  local marker expansion
- 14:  $P_i \leftarrow G \wedge S_i^+$   $\triangleright$  recover pane pixels from original glass mask
- 15:  $P_i \leftarrow P_i \wedge \neg F$   $\triangleright$  remove only original frame pixels
- 16:  $C_i \leftarrow \text{LARGESTCONTOUR}(P_i)$
- 17:  $Q_i \leftarrow \text{ORDERCORNERS}(\text{BOXPOINTS}(\text{MINAREARECT}(C_i)))$
- 18:  $Q \leftarrow Q \cup \{Q_i\}$
- 19: **end for**
- 20: **if**  $Q = \emptyset$  **then**
- 21: **fail**  $\triangleright$  no valid panel instances found
- 22: **end if**
- 23:  $Q_{\text{ref}} \leftarrow \text{SELECTTOPRIGHT}(Q)$   $\triangleright$  top-most row, then right-most
- 24: **return**  $Q, Q_{\text{ref}}$

contains 571 high-resolution images with diverse resolutions ranging from  $612 \times 144$  to  $3888 \times 592$ .

While the real dataset is sufficient for robust facade segmentation, the lack of ground truth poses affects the downstream qualitative pose estimation evaluation of the proposed method. To address these constraints, we generated a synthetic dataset in a MuJoCo (Todorov et al., 2012). The simulated scene is based on a 1:25 scale building prototype shown in Fig. 7. Unitized CWM (Fig. 7d) measures 153 mm in height and 103 mm in width and consists of float glass within an aluminum frame. To generate diverse facade configurations, we began with a complete building facade and created seven structural variants by systematically removing panels from different sides. For each variant, additional panels were randomly removed during rendering to simulate a partially installed building and to enhance layout variety. Scenes were rendered from multiple virtual camera angles under four lighting conditions. For every combination of configuration, camera, lighting, we exported an RGB image and its semantic mask at a fixed resolution of  $384 \times 384$ . The resulting synthetic set contains 5,622 image-mask pairs, with example samples shown in Fig. 5c. In this paper, we treat the real and synthetic subsets as a unified dataset comprising 6,193 images for training LAnet and conducting pose-estimation experiments. The dataset comprises 571 real images and 5,622 synthetic images and is randomly split into training and validation sets with an 80/20 partition. Notably, the pixel-level class distribution is inherently imbalanced (Fig. 6): the *frame* class accounts for only 10% of pixels, whereas *glass* and *background* each account for approximately 45%. This indicates that thin-frame struc-

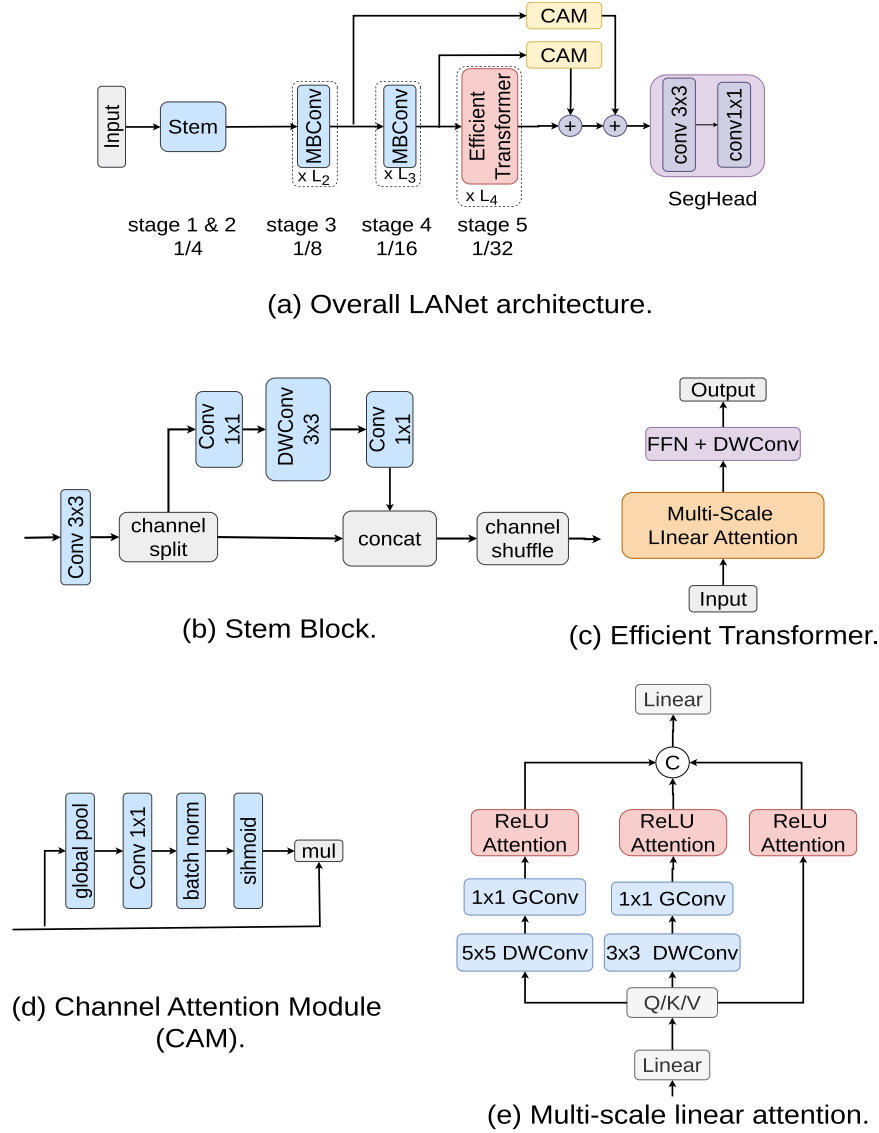


Fig. 4. LANet semantic segmentation model (Hassen et al., 2025).

tures are underrepresented relative to large glass and background regions. This imbalance motivates our instance-separation strategy and geometry-based post-processing, which are designed to preserve thin frame boundaries for reliable panel extraction.

## 4. Experiments

### 4.1. Simulation environment

To validate the proposed perception-to-pose pipeline for estimating the pose of CWM target installation slots, we implement a physics-based MuJoCo simulation of a building prototype and CWMs, as shown in Fig. 7. A virtual RGB camera is fixed in the scene with respect to the world frame, enabling direct comparison of the estimated 6D pose with the ground truth.

### 4.2. Segmentation implementation details

For LANet training, we follow the protocol in Hassen et al. (2025). All experiments were implemented using MMSegmentation (MMSEG, 2020) and trained on an NVIDIA RTX A6000 GPU. The batch size was set to 16, and all input images were resized to  $384 \times 384$ . We trained

Table 1

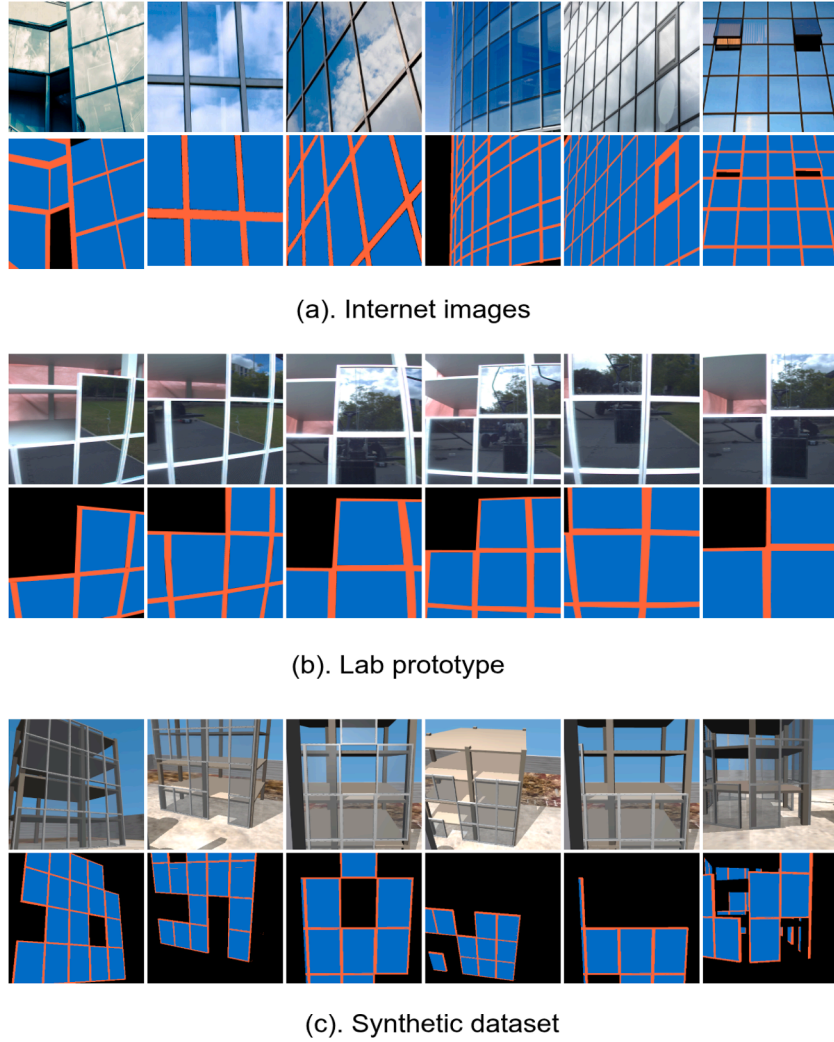
Per-class segmentation performance of LANet on the proposed dataset.

| Class       | IoU (%)      | Accuracy (%) |
|-------------|--------------|--------------|
| Frame       | 69.27        | 85.43        |
| Glass       | 91.35        | 96.28        |
| Background  | 93.10        | 94.75        |
| <b>Mean</b> | <b>84.57</b> | <b>92.15</b> |

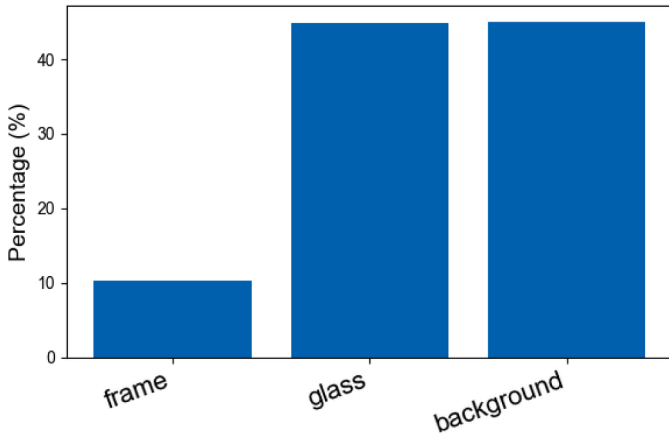
the network for 100 epochs using the AdamW optimizer with an initial learning rate of  $1 \times 10^{-4}$  and a weight decay of 0.05. We used a polynomial learning-rate decay schedule with power 0.9. For training and validation, the dataset described in the previous section was used. At the same time, the pose-estimation stage was evaluated on real-time RGB images rendered from a MuJoCo physics-based simulation.

### 4.3. Evaluation metrics

The proposed framework is *modular*, comprising semantic segmentation (Stage 1) and 6D pose estimation (Stage 2). Since the pose-estimation module produces different outputs and is evaluated by differ-



**Fig. 5.** Representative samples from the unified curtain-wall dataset. (a) In-the-wild internet images. (b) Lab-scale prototype images. (c) MuJoCo-generated synthetic images. For each subset, the top row shows RGB inputs and the bottom row shows the annotated segmentation masks (*frame* in orange, *glass* in blue, *background* in black).



**Fig. 6.** Pixel-level class distribution (%) of the unified dataset across the three semantic classes (frame, glass, background), showing that the frame class is a minority compared with the dominant glass and background regions.

ent error criteria than semantic segmentation, we evaluate each stage using task-appropriate metrics. Specifically, to quantify segmentation

accuracy and runtime efficiency, we use mean Intersection-over-Union (mIoU) and pixel accuracy, whereas for pose estimation, we report translational and orientation errors.

Let  $TP_c$ ,  $FP_c$ ,  $FN_c$ , and  $TN_c$  denote the number of true positive, false positive, false negative, and true negative pixels for class  $c \in \{frame, glass, background\}$ , respectively. Here,  $TP_c$  corresponds to pixels correctly predicted as class  $c$ ;  $FP_c$  denotes pixels that do not belong to class  $c$  but are incorrectly predicted as  $c$ ;  $FN_c$  denotes pixels of class  $c$  that are incorrectly predicted as another class; and  $TN_c$  denotes pixels correctly predicted as not belonging to class  $c$ .

The metrics are defined as

$$IoU_c = \frac{TP_c}{TP_c + FP_c + FN_c}, \quad (4)$$

$$mIoU = \frac{1}{|C|} \sum_{c \in C} IoU_c, \quad C = \{frame, glass, background\}, \quad (5)$$

$$Accuracy = \frac{\sum_{c \in C} TP_c}{\sum_{c \in C} (TP_c + FN_c)}. \quad (6)$$

where  $TP_c$ ,  $FP_c$ , and  $FN_c$  denote the true positive, false positive, and false negative pixel counts for class  $c$ , respectively. Here,  $IoU_c$  measures the overlap between the predicted and ground-truth regions for class  $c$ , and mIoU reports the average overlap across the three classes. Pixel

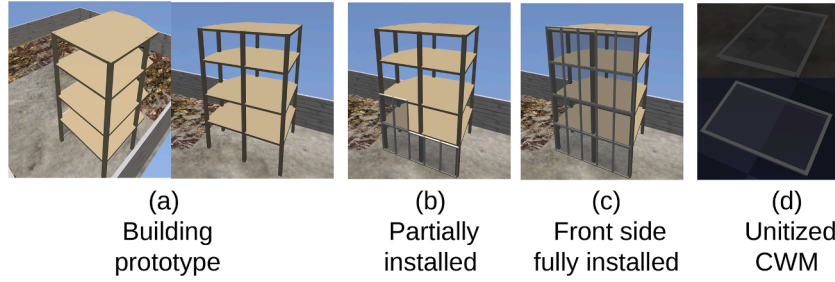


Fig. 7. Virtual building prototype in MuJoCo.

Table 2

Ablation of OpenCV PnP solvers for the second-stage pose estimation. Lower values are better for errors and runtime; higher values are better for accuracy.

| PnP method                        | Trans. error (mm) ↓ | Rot. error (°) ↓ | Reproj. error (px) ↓ | Runtime (ms) ↓ |
|-----------------------------------|---------------------|------------------|----------------------|----------------|
| SOLVEPNP_ITERATIVE                | 6.709               | 10.156           | 0.053                | 0.311          |
| SOLVEPNP_EPNP                     | 6.958               | 10.000           | 0.071                | 0.152          |
| SOLVEPNP_IPPE                     | 6.478               | 10.374           | 0.080                | 0.187          |
| SOLVEPNP_SQPNP                    | 6.709               | 10.156           | 0.053                | 0.214          |
| solvePnP Ransac (EPNP) + RefineLM | 6.709               | 10.156           | 0.053                | 0.244          |
| solvePnP Ransac (AP3P) + RefineLM | 6.709               | 10.156           | 0.053                | 0.297          |

Accuracy measures the proportion of correctly classified pixels over the entire image.

To evaluate downstream pose estimation, translation accuracy is quantified using (i) the full 3D Euclidean error.

$$d' = \|\hat{\mathbf{p}} - \mathbf{p}_{gt}\|_2, \quad (7)$$

and (ii) the in-plane (XY) error that ignores depth, to reflect practical lateral alignment tolerances.

$$d'' = \|\hat{\mathbf{x}} - \mathbf{x}_{gt}, \hat{\mathbf{y}} - \mathbf{y}_{gt}\|_2, \quad (8)$$

Orientation accuracy is measured using the quaternion geodesic distance

$$e_\theta = 2 \cos^{-1} \left( \left| \langle \hat{\mathbf{q}}, \mathbf{q}_{gt} \rangle \right| \right), \quad (9)$$

reported in degrees, where all quaternions are unit-normalized and expressed in  $(w, x, y, z)$  order.

## 5. Results

### 5.1. Segmentation result

This subsection presents the quantitative and qualitative performance of LANet on our custom dataset.

**Quantitative results.** Table 1 reports the per-class Intersection-over-Union (IoU) and accuracy achieved by LANet on our dataset. The model achieves strong overall performance, with an mIoU of 84.57% and a mean class accuracy (mAcc) of 92.15%. Performance is highest on *Background* (IoU 93.10%) and *Glass* (IoU 91.35%), indicating reliable separation of large facade regions and transparent panel areas under diverse appearance conditions. The *Frame* class remains the most challenging (IoU 69.27%), which is expected given its thin structures, frequent occlusions, and strong inherited class imbalance relative to *glass* and *background*. Importantly, despite the lower frame IoU, the achieved accuracy for *Frame* (85.43%) indicates that most frame pixels are correctly identified, with residual errors primarily near boundaries. Overall, these results demonstrate that LANet provides sufficiently accurate facade primitives to support downstream geometric reasoning and pose estimation in CWM installation scenarios.

**Qualitative results.** Fig. 8 shows qualitative segmentation results of LANet, comparing the input image, ground-truth mask, and predicted mask for (a) synthetic samples and (b) real samples. Overall, LANet produces visually consistent facade parsing, with *glass* regions recovered as

contiguous planar areas and *frame* structures delineated with largely correct topology across both synthetic and real images. The predictions preserve the global curtain-wall grid pattern across varying viewpoints and lighting conditions. Most residual errors occur along thin *frame* boundaries and at high-curvature perspective regions, where small misalignments lead to visible boundary shifts.

### 5.2. Pose estimation results

We evaluate reference-panel and target slot pose estimation using the downstream pose metrics defined in Section 4.3. All pose-estimation experiments use a fixed pinhole camera model with the intrinsic matrix

$$K = \begin{bmatrix} 463.5290 & 0 & 192 \\ 0 & 463.5290 & 192 \\ 0 & 0 & 1 \end{bmatrix}, \quad (10)$$

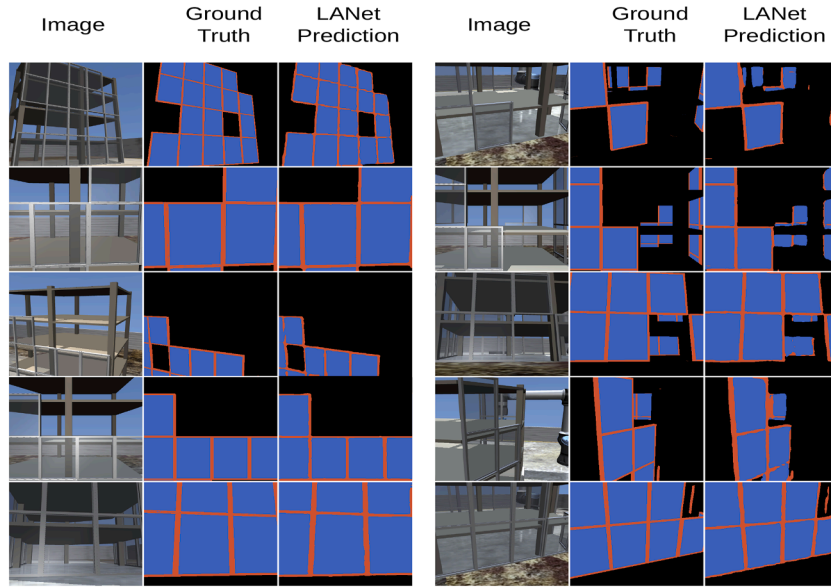
corresponding to a  $384 \times 384$  image with principal point at  $(c_x, c_y) = (192, 192)$  pixels and  $f_x = f_y = 463.5290$  pixels. We report pose errors in the camera coordinate frame. Translation accuracy is reported as the full 3D Euclidean error  $d'$  and the in-plane error  $d''$  (XY only), while orientation accuracy is reported as the quaternion geodesic distance  $e_\theta$  error.

**Ablation study on the PnP methods.** We conducted an ablation study across different PnP solvers to quantify the impact of OpenCV PnP variants on downstream pose accuracy and efficiency. Specifically, we evaluated SOLVEPNP\_ITERATIVE, SOLVEPNP\_EPNP, SOLVEPNP\_IPPE, SOLVEPNP\_SQPNP, and robust variants using solvePnP Ransac combined with Levenberg–Marquardt refinement (RefineLM). We report translation error (mm), quaternion geodesic rotation error (degrees), reprojection RMSE (pixels), and runtime (ms).

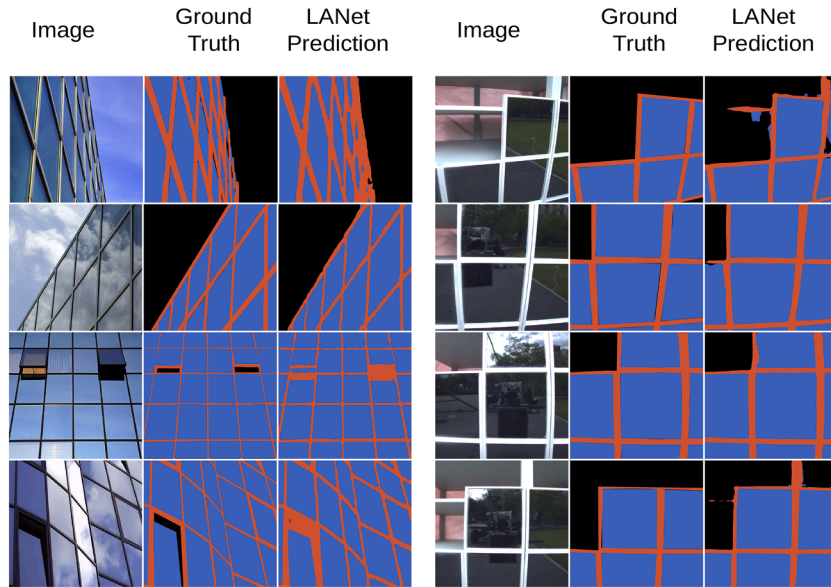
As shown in Table 2, SOLVEPNP\_IPPE achieves the lowest translation error (6.478 mm), which is consistent with our planar target geometry (rectangle corners), where IPPE is designed to provide stable pose estimates. Meanwhile, SOLVEPNP\_ITERATIVE and SOLVEPNP\_SQPNP produce the lowest reprojection error (0.053 px), but with slightly higher translation error. Among the evaluated methods, SOLVEPNP\_EPNP achieves the fastest runtime (0.152 ms) but yields slightly lower pose accuracy than IPPE. The RANSAC-based variants do not improve results in this setting, as expected, because the pose is estimated from four ordered corners (i.e., no redundant correspondences for outlier rejection).

Based on this ablation, we adopt SOLVEPNP\_IPPE as the default solver in our pipeline. If IPPE fails due to numerical/degenerate configurations,





(a) Qualitative segmentation results of LANet on synthetic samples.



(b) Qualitative segmentation results of LANet on real samples.

**Fig. 8.** Qualitative segmentation results of LANet on representative samples. For each example, we show the input image, the ground-truth mask, and the LANet prediction. LANet accurately recovers the curtain-wall layout and glass regions, while most errors are confined to thin frame boundaries and challenging lighting or perspective conditions.

it falls back to SOLVEPNP\_ITERATIVE as a reliable alternative. This choice provides a good balance between stability on planar targets and consistent downstream pose accuracy.

**Baseline definition and comparison.** Currently, there is no publicly available benchmark dataset or reference codebase specifically for estimating the pose of CWMs target installation slots. Therefore, we adopt a marker-based AprilTag3 (Krogius et al., 2019) as the baseline. This choice is motivated by two practical considerations: (i) marker-based pose estimation is a widely accepted and reproducible geometric reference in robotics, providing a strong reference baseline for pose accuracy, and (ii) it enables a fair comparison under identical camera viewpoints and evaluation protocols. For the baseline, we placed a square April-

Tag3 of size 25 mm × 25 mm at the geometric center of the reference panel and estimated the reference panel poses from the markers. We evaluate reference-panel pose for 15 panels across three yaw camera configurations, yielding 45 cases, as shown in Table A.2. For the target slot pose, both the proposed method and the baseline use the same deterministic pose propagation from the estimated reference panel pose. The target slot is not defined for reference panels located at the row boundary (i.e., the last/rightmost panel in the evaluated current installation row) because there is no subsequent installation slot within the modeled scene. Consequently, these boundary panels are excluded from the target slot pose evaluation, leaving 12 valid installation target slots, yielding 36 cases.

Table 3

Error comparison of reference-panel and target-slot pose estimation across yaw scenarios (proposed vs AprilTag).

| Scenario | Yaw  | Reference panel |               | Orientation error       |                         | Target slot   |               |
|----------|------|-----------------|---------------|-------------------------|-------------------------|---------------|---------------|
|          |      | Proposed(mm)    | Apriltag (mm) | Proposed ( $e_\theta$ ) | Apriltag ( $e_\theta$ ) | Proposed (mm) | Apriltag (mm) |
| 1        | 0°   | 9.27            | <b>1.41</b>   | <b>8.33</b>             | 8.39                    | <b>1.41</b>   | 9.27          |
|          | +5°  | 9.22            | <b>1.00</b>   | 2.75                    | <b>1.66</b>             | 11.09         | <b>3.16</b>   |
|          | +10° | 14.87           | <b>4.58</b>   | 7.39                    | <b>3.55</b>             | 27.24         | <b>7.07</b>   |
| 2        | 0°   | 10.10           | <b>1.00</b>   | <b>7.01</b>             | 15.97                   | <b>10.25</b>  | 12.21         |
|          | +5°  | 8.06            | <b>6.08</b>   | 4.56                    | <b>3.37</b>             | 16.12         | <b>12.00</b>  |
|          | +10° | <b>6.16</b>     | 7.07          | 10.37                   | <b>5.57</b>             | 25.02         | <b>8.31</b>   |
| 3        | 0°   | 7.62            | <b>7.14</b>   | 2.92                    | <b>1.03</b>             | 10.82         | <b>5.48</b>   |
|          | +5°  | <b>4.36</b>     | 6.32          | 5.38                    | <b>3.03</b>             | 3.00          | <b>2.24</b>   |
|          | +10° | 17.92           | <b>13.49</b>  | <b>9.76</b>             | 10.39                   | 13.93         | <b>7.28</b>   |
| 4        | 0°   | 11.45           | <b>8.49</b>   | 17.19                   | <b>0.26</b>             | 11.18         | <b>8.31</b>   |
|          | +5°  | 16.16           | <b>7.35</b>   | 17.55                   | <b>3.74</b>             | 44.11         | <b>14.07</b>  |
|          | +10° | 14.07           | <b>5.10</b>   | <b>15.43</b>            | 28.88                   | 40.46         | <b>37.62</b>  |
| 5        | 0°   | <b>3.16</b>     | 8.49          | 0.99                    | <b>0.11</b>             | —             | —             |
|          | +5°  | 51.36           | <b>50.05</b>  | <b>4.88</b>             | 9.66                    | —             | —             |
|          | +10° | 9.64            | <b>1.00</b>   | 9.10                    | <b>7.12</b>             | —             | —             |
| 6        | 0°   | 7.14            | <b>0.00</b>   | <b>2.98</b>             | 7.24                    | <b>4.24</b>   | 10.10         |
|          | +5°  | 4.24            | <b>3.16</b>   | 3.18                    | <b>2.78</b>             | 7.07          | <b>2.45</b>   |
|          | +10° | 14.35           | <b>5.48</b>   | <b>6.84</b>             | 10.35                   | 26.25         | <b>11.45</b>  |
| 7        | 0°   | 9.11            | <b>2.00</b>   | 2.12                    | <b>0.77</b>             | 9.43          | <b>1.00</b>   |
|          | +5°  | <b>4.12</b>     | 6.08          | 8.50                    | <b>3.22</b>             | 17.15         | <b>12.04</b>  |
|          | +10° | 11.36           | <b>6.08</b>   | 21.31                   | <b>8.71</b>             | 43.16         | <b>15.07</b>  |
| 8        | 0°   | 7.62            | <b>7.35</b>   | <b>2.40</b>             | 7.25                    | 10.82         | <b>5.92</b>   |
|          | +5°  | 6.32            | <b>5.20</b>   | <b>1.37</b>             | 3.24                    | 7.35          | <b>2.24</b>   |
|          | +10° | 17.49           | <b>3.00</b>   | 4.34                    | <b>2.40</b>             | 14.04         | <b>2.45</b>   |
| 9        | 0°   | <b>1.41</b>     | 8.12          | 8.29                    | <b>4.30</b>             | 7.28          | <b>7.14</b>   |
|          | +5°  | 7.07            | <b>5.10</b>   | 2.82                    | <b>4.49</b>             | <b>12.00</b>  | 13.19         |
|          | +10° | 15.33           | <b>6.08</b>   | 20.63                   | <b>19.26</b>            | 52.24         | <b>43.05</b>  |
| 10       | 0°   | 11.75           | <b>4.12</b>   | 1.64                    | 3.12                    | —             | —             |
|          | +5°  | 8.06            | <b>6.08</b>   | <b>4.91</b>             | 5.23                    | —             | —             |
|          | +10° | <b>9.70</b>     | 15.13         | <b>2.83</b>             | 33.19                   | —             | —             |
| 11       | 0°   | <b>2.00</b>     | 5.10          | 11.21                   | <b>2.61</b>             | 16.55         | <b>2.24</b>   |
|          | +5°  | <b>3.16</b>     | 4.12          | <b>1.34</b>             | 2.72                    | <b>4.00</b>   | 7.07          |
|          | +10° | <b>3.16</b>     | 11.45         | <b>2.73</b>             | 7.79                    | <b>3.00</b>   | 24.35         |
| 12       | 0°   | <b>2.45</b>     | 8.06          | <b>6.31</b>             | 9.42                    | <b>2.45</b>   | 9.22          |
|          | +5°  | 7.07            | <b>6.08</b>   | 9.55                    | <b>4.85</b>             | 24.10         | <b>15.03</b>  |
|          | +10° | 11.09           | <b>8.06</b>   | 18.19                   | <b>1.29</b>             | 43.05         | <b>11.05</b>  |
| 13       | 0°   | <b>6.08</b>     | 7.35          | 2.75                    | <b>1.31</b>             | 2.24          | 5.39          |
|          | +5°  | <b>3.74</b>     | 12.21         | <b>2.48</b>             | 5.74                    | <b>2.45</b>   | 3.61          |
|          | +10° | <b>7.07</b>     | 10.10         | <b>7.68</b>             | 11.56                   | 20.00         | <b>10.95</b>  |
| 14       | 0°   | <b>2.00</b>     | 8.06          | 1.33                    | <b>0.77</b>             | <b>1.41</b>   | 8.06          |
|          | +5°  | 8.12            | <b>7.14</b>   | 13.47                   | <b>1.39</b>             | 31.14         | <b>4.00</b>   |
|          | +10° | <b>6.32</b>     | 15.56         | 3.89                    | <b>7.93</b>             | <b>12.21</b>  | 28.46         |
| 15       | 0°   | <b>6.00</b>     | 10.10         | <b>0.70</b>             | 2.12                    | —             | —             |
|          | +5°  | <b>2.45</b>     | 4.00          | 4.02                    | <b>1.39</b>             | —             | —             |
|          | +10° | <b>8.25</b>     | 28.46         | 20.12                   | <b>7.93</b>             | —             | —             |
| Mean     |      | 9.05            | 7.94          | 7.19                    | 6.38                    | 16.34         | 10.63         |
| Median   |      | 7.62            | 6.08          | 4.91                    | 4.30                    | 11.59         | 8.31          |

**Camera-frame convention.** In this work, PnP is first solved in the standard OpenCV camera frame, where +x points to the right in the image, +y points downward, and +z points forward. For consistency with MuJoCo, the estimated pose is then expressed in the MuJoCo camera frame using the fixed axis conversion  $C_{cv \rightarrow mj} = \text{diag}(1, -1, -1)$ :

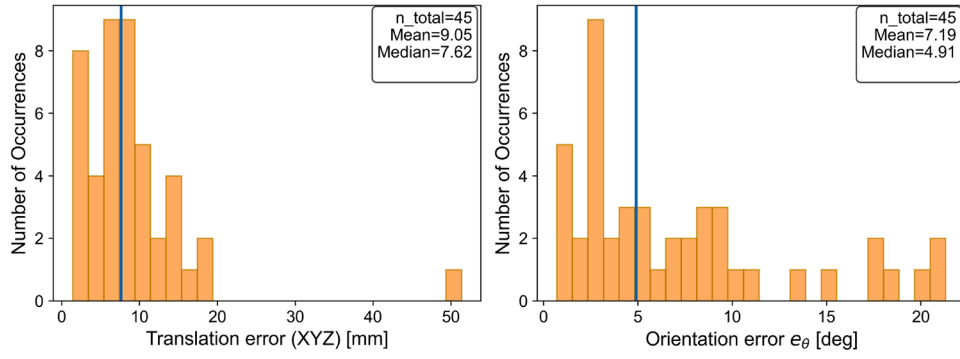
$$R_{co}^{mj} = C_{cv \rightarrow mj} R_{co}^{cv} C_{cv \rightarrow mj}^T, \quad t_{co}^{mj} = C_{cv \rightarrow mj} t_{co}^{cv}. \quad (11)$$

Under this convention, +x remains right, +y becomes up, and points in front of the camera have  $z < 0$  in the MuJoCo camera frame (e.g.,  $z \approx -500$  mm for a target at  $\approx 0.5$  m range).

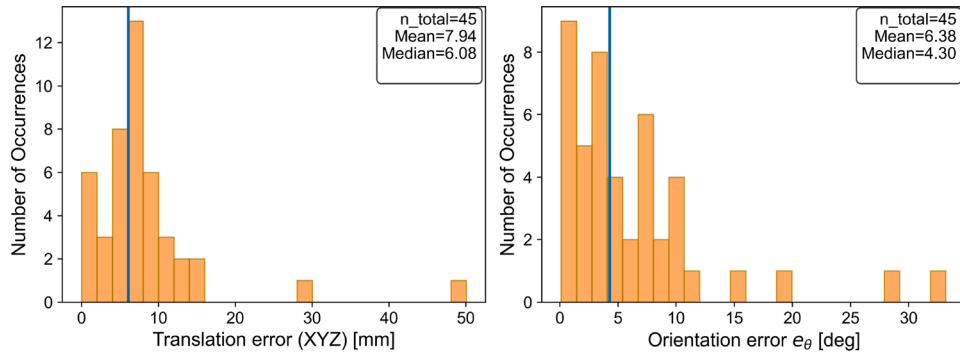
**Reference-panel pose estimation.** Across the 45 reference-panel pose test cases, the proposed pipeline achieves a median translation error of 7.62 mm, compared with 6.08 mm for the AprilTag baseline (Table 3). Overall, the proposed method remains stable for most panels and viewpoints, with errors concentrated in the single-digit millimeter range; however, a small number of large-error outliers are observed under challenging viewpoints, indicating occasional failures. Fig. 11 qualitatively shows typical failed cases and how segmentation errors affect pose estimation. Specifically, incomplete frame separation can cause neighboring glass regions to adhere, resulting in merged instances and a large quadrilateral spanning multiple panels; conversely, fragmented or partially missing glass boundaries can bias the fitted quadrilateral, thereby shifting the recovered corner positions. These corner deviations directly impact the PnP stage through (i) corner displacement (higher reprojection residuals and pose bias) and (ii) correspondence failure (missing or degenerate quadrilaterals or inconsistent corner ordering), which may lead to unstable or invalid pose solutions. Furthermore, boundary and high-yaw viewpoints convert small corner localization errors into depth-dominant translation errors, which explains the outliers in the reference-

pose distribution. To stop unreliable estimates from affecting subsequent steps, we use a reprojection-based validity check and reject pose estimates when the reprojection RMSE exceeds 6 pixels (Figure 3 3). Orientation errors are evaluated on the 45 reference cases. The proposed method achieves a median angular error of 4.91°, compared with 4.30° for AprilTag (Table 3). In terms of robustness, 77.8% of proposed estimates fall within 10° (vs. 84.4% for AprilTag), indicating broadly competitive orientation performance with a modest advantage to the marker baseline.

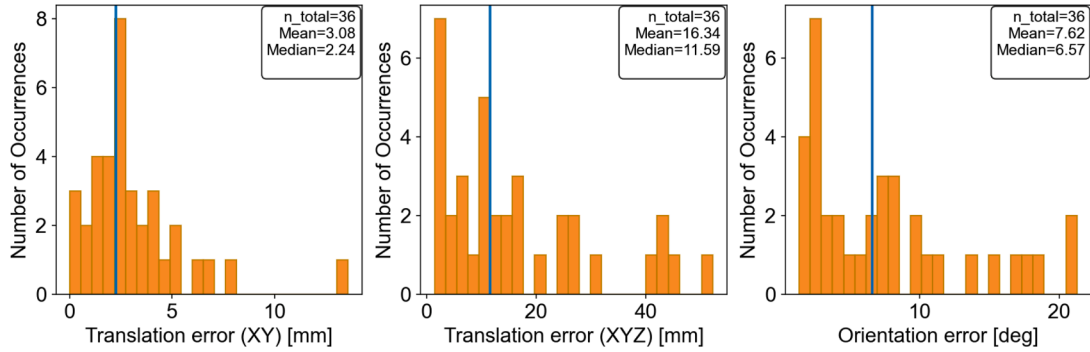
**Target slot pose estimation.** Target slot pose results are reported over 36 test cases in Table 3. In XYZ error ( $d'$ ), the proposed method attains a median target translation error of 11.59 mm, whereas the AprilTag baseline achieves 8.31 mm. In contrast, the proposed method provides strong in-plane localization: the XY error ( $d''$ ) is tightly concentrated (median 2.24 mm; Table A.2), indicating reliable lateral placement. The gap between  $d''$  and  $d'$  indicates that the residual error is dominated by depth ( $Z$ ) uncertainty along the viewing direction, consistent with the right-tailed  $d'$  distributions in Figs. 9 and 10. This behavior is most evident under oblique viewpoints, where small segmentation drift at thin frame boundaries can induce larger depth deviations due to the PnP algorithm. In Figs. 9 and 10, the y-axis label “Number of occurrences” denotes the histogram frequency (i.e., the number of test samples falling within each bin). The legend reports summary statistics of the plotted distribution, including the number of test cases ( $n$ ), the mean, and the median error. Overall, the proposed markerless pipeline provides accurate in-plane target-slot localization and moderate orientation accuracy, whereas full 3D target-slot accuracy is primarily limited by depth sensitivity under oblique viewpoints.



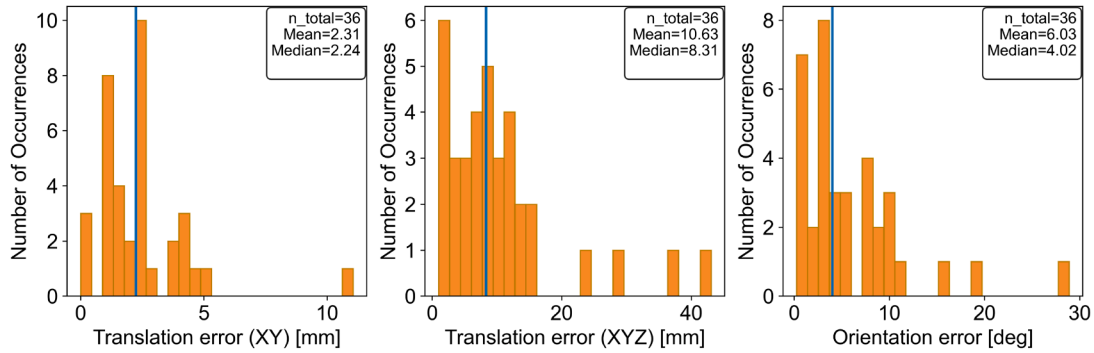
(a) Proposed method error.



(b) Markers method error.

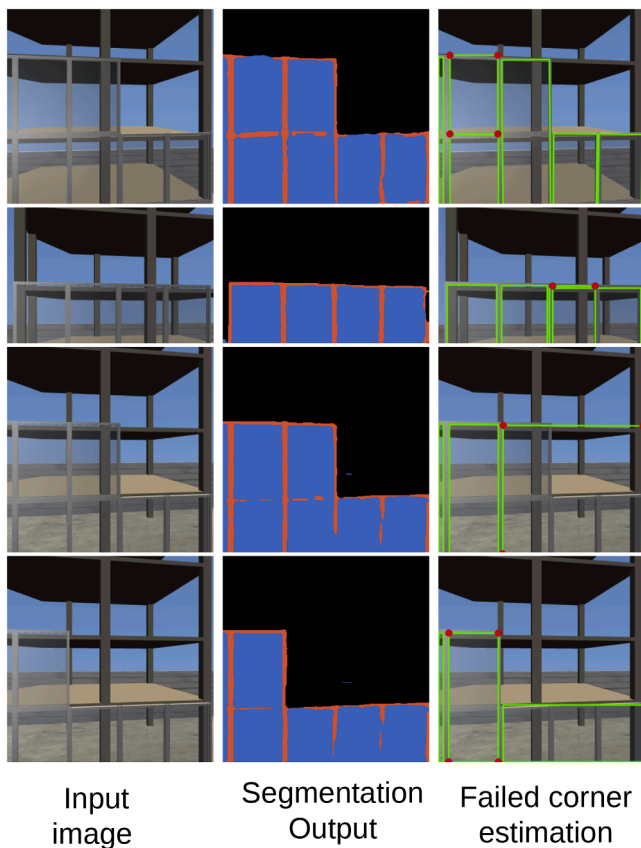
**Fig. 9.** Reference-panel pose estimation error distributions across 45 test cases.

(a) Proposed method error.



(b) Markers method error.

**Fig. 10.** Distribution of target-slot pose errors across 36 target-slot test cases.



**Fig. 11.** Qualitative failure cases of the proposed markerless pipeline. Each column shows (left) the input RGB image, (middle) the predicted semantic mask, and (right) the extracted quadrilateral and corner candidates used for planar PnP.

### 5.3. Academic and practical implications

This study contributes to construction robotics by formalizing a markerless, monocular 6D pose-estimation pipeline tailored to curtain-wall installation. It demonstrates how lightweight semantic segmentation can be coupled with geometric reasoning to bridge perception and manipulation in a computationally efficient manner. The work also highlights a broader research direction: moving beyond detection/segmentation studies toward pose-aware, control-oriented perception.

For practitioners and project managers, the proposed approach can enable safer, more repeatable installation workflows by reducing reliance on manual alignment and crane-signal coordination, which are major sources of on-site risk and variability. A lightweight markerless perception stack can also reduce operational overhead relative to marker-based systems (e.g., tag placement and visibility constraints) while still producing pose estimates suitable for closed-loop alignment. In the longer term, integrating pose estimation with compliant control can improve placement consistency, reduce rework, and strengthen quality assurance, particularly in constrained or high-risk working-at-height conditions.

## 6. Conclusion, limitations and future work

This study presented a markerless, vision-based approach for localizing and estimating the 6D pose of a CWM target installation slot to

support autonomous robotic alignment in CWM installation. The proposed pipeline combines a lightweight semantic segmentation network with deterministic geometric reasoning to recover an ordered set of panel corners from facade images, enabling pose recovery through a PnP solver. The reference installed-panel pose is subsequently propagated along the panel local width axis to obtain the target slot pose.

Quantitative evaluation in a physics-based simulation environment demonstrated that the proposed method can localize the target slot translation pose with millimeter-to-centimeter accuracy, achieving a median 3D translation error of 11.59 mm (mean 16.34 mm), a median planar error of 2.24 mm (mean 3.08 mm), and a median orientation error of 4.91° (mean 7.19°) under the tested yaw variations. The proposed markerless pipeline remains attractive for construction scenarios where deploying markers is undesirable, impractical, or unsafe, and where a segmentation-driven perception module can be integrated directly into robotic alignment workflows.

**Limitations and future work:** Despite these promising results, several limitations should be addressed. First, because pose estimation is derived from monocular corner correspondences, depth-related components of the translation error can be sensitive to pixel-level corner localization noise, particularly under oblique viewpoints. In future work, depth stability can be improved by (a) fusing multi-view observations over short camera motions, (b) incorporating RGB-D sensing to constrain the Z component directly, and (c) leveraging scene priors—such as curtain-wall planarity and known module spacing—within a constrained optimization that minimizes reprojection error while regularizing the pose to satisfy planar/layout constraints. Second, all pose-accuracy results reported in this paper are validated in simulation. A real-world pose validation remains future work and is essential before field deployment. Future work will therefore prioritize: (i) collecting large-scale real-site or lab-scale facade datasets with reliable ground-truth pose and (ii) integrating the estimated target-slot pose into a closed-loop robotic CWM installation system.

### CRedit authorship contribution statement

**Ahmed Yimam Hassen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization; **Mehrdad Arashpour:** Writing – review & editing, Visualization, Formal analysis, Validation, Supervision; **Elahe Abdi:** Writing – review & editing, Visualization, Validation, Formal analysis, Supervision, Resources, Project administration, Investigation, Funding acquisition.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

### Appendix A. Additional pose results

This appendix reports additional per-case pose-estimation results (Tables A.1–A.3) that complement the aggregate statistics presented in the main text.

**Table A.1**

Translational pose-estimation error for the reference panel: proposed method versus AprilTag baseline.

| Scenario | Ground truth |       |       | Proposed |       |       | $d'$  | AprilTag |       |       | $d'$  |
|----------|--------------|-------|-------|----------|-------|-------|-------|----------|-------|-------|-------|
|          | X(mm)        | Y(mm) | Z(mm) | X(mm)    | Y(mm) | Z(mm) |       | X(mm)    | Y(mm) | Z(mm) |       |
| 1        | -99          | -70   | -500  | -97      | -69   | -491  | 9.27  | -98      | -70   | -499  | 1.41  |
|          | -142         | -70   | -490  | -140     | -70   | -481  | 9.22  | -142     | -70   | -489  | 1.00  |
|          | -135         | -90   | -484  | -132     | -86   | -470  | 14.87 | -133     | -89   | -480  | 4.58  |
| 2        | 5            | -70   | -500  | 4        | -69   | -490  | 10.10 | 5        | -70   | -501  | 1.00  |
|          | -39          | -70   | -498  | -40      | -70   | -490  | 8.06  | -39      | -69   | -492  | 6.08  |
|          | -33          | -70   | -502  | -34      | -69   | -496  | 6.16  | -33      | -69   | -495  | 7.07  |
| 3        | 108          | -120  | -500  | 105      | -120  | -493  | 7.62  | 106      | -119  | -493  | 7.14  |
|          | 64           | -120  | -507  | 62       | -121  | -510  | 4.36  | 64       | -122  | -513  | 6.32  |
|          | 68           | -120  | -520  | 64       | -116  | -503  | 17.92 | 66       | -117  | -507  | 13.49 |
| 4        | 11           | -120  | -500  | 10       | -113  | -493  | 11.45 | 10       | -118  | -492  | 8.49  |
|          | -33          | -120  | -499  | -34      | -112  | -485  | 16.16 | -32      | -118  | -492  | 7.35  |
|          | -27          | -120  | -503  | -29      | -115  | -490  | 14.07 | -27      | -119  | -498  | 5.10  |
| 5        | 114          | -120  | -500  | 113      | -120  | -497  | 3.16  | 112      | -118  | -492  | 8.49  |
|          | 70           | -120  | -508  | 19       | -121  | -502  | 51.36 | 20       | -121  | -506  | 50.05 |
|          | 74           | -120  | -521  | 72       | -115  | -513  | 9.64  | 74       | -120  | -520  | 1.00  |
| 6        | -99          | -67   | -500  | -98      | -66   | -493  | 7.14  | -99      | -67   | -500  | 0.00  |
|          | -142         | -67   | -490  | -141     | -66   | -486  | 4.24  | -141     | -67   | -487  | 3.16  |
|          | -135         | -67   | -484  | -132     | -66   | -470  | 14.35 | -133     | -66   | -479  | 5.48  |
| 7        | 108          | -67   | -500  | 105      | -67   | -493  | 7.62  | 106      | -66   | -493  | 7.35  |
|          | 64           | -67   | -507  | 62       | -67   | -501  | 6.32  | 63       | -66   | -502  | 5.20  |
|          | 68           | -67   | -520  | 64       | -66   | -503  | 17.49 | 68       | -67   | -517  | 3.00  |
| 8        | 5            | -67   | -500  | 4        | -66   | -491  | 9.11  | 5        | -67   | -498  | 2.00  |
|          | -39          | -67   | -498  | -40      | -67   | -494  | 4.12  | -39      | -66   | -492  | 6.08  |
|          | -33          | -67   | -502  | -35      | -65   | -491  | 11.36 | -33      | -66   | -496  | 6.08  |
| 9        | 11           | -67   | -500  | 11       | -66   | -499  | 1.41  | 10       | -66   | -492  | 8.12  |
|          | -83          | -67   | -495  | -82      | -67   | -488  | 7.07  | -82      | -66   | -490  | 5.10  |
|          | -27          | -67   | -503  | -28      | -64   | -488  | 15.33 | -27      | -66   | -497  | 6.08  |
| 10       | 114          | -67   | -500  | 110      | -66   | -489  | 11.75 | 113      | -67   | -496  | 4.12  |
|          | 20           | -67   | -504  | 19       | -67   | -496  | 8.06  | 20       | -66   | -498  | 6.08  |
|          | 74           | -67   | -521  | 72       | -64   | -513  | 9.70  | 72       | -65   | -506  | 15.13 |
| 11       | -99          | -14   | -500  | -99      | -16   | -500  | 2.00  | -98      | -14   | -495  | 5.10  |
|          | -142         | -14   | -490  | -141     | -14   | -487  | 3.16  | -141     | -14   | -486  | 4.12  |
|          | -135         | -14   | -484  | -136     | -14   | -487  | 3.16  | -132     | -13   | -473  | 11.45 |
| 12       | 5            | -14   | -500  | 4        | -15   | -502  | 2.45  | 5        | -13   | -492  | 8.06  |
|          | -39          | -14   | -498  | -40      | -14   | -491  | 7.07  | -39      | -13   | -492  | 6.08  |
|          | -33          | -14   | -502  | -35      | -15   | -491  | 11.09 | -33      | -13   | -494  | 8.06  |
| 13       | 108          | -14   | -500  | 107      | -14   | -506  | 6.08  | 106      | -13   | -493  | 7.35  |
|          | 64           | -14   | -507  | 62       | -15   | -510  | 3.74  | 62       | -13   | -495  | 12.21 |
|          | -79          | -14   | -494  | -80      | -14   | -487  | 7.07  | -78      | -13   | -484  | 10.10 |
| 14       | -39          | -14   | -500  | -39      | -14   | -502  | 2.00  | -39      | -13   | -492  | 8.06  |
|          | -33          | -14   | -499  | -34      | -15   | -491  | 8.12  | -32      | -13   | -492  | 7.14  |
|          | -126         | -14   | -486  | -124     | -14   | -480  | 6.32  | -122     | -13   | -471  | 15.56 |
| 15       | 114          | -14   | -500  | 114      | -14   | -506  | 6.00  | 113      | -13   | -490  | 10.10 |
|          | 70           | -14   | -508  | 69       | -15   | -506  | 2.45  | 70       | -14   | -504  | 4.00  |
|          | -24          | -14   | -503  | -26      | -14   | -495  | 8.25  | -19      | -13   | -475  | 28.46 |
| Mean     |              |       |       |          |       |       | 9.05  |          |       |       | 7.94  |
| Median   |              |       |       |          |       |       | 7.62  |          |       |       | 6.08  |



**Table A.2**

Translational pose-estimation error for the target installation slot: proposed method versus AprilTag baseline.

| Scenario      | Ground truth |       |       | Proposed |       |       | $d'$ (mm)    | $d''$ (mm)  | AprilTag |       |       | $d'$ (mm)    | $d''$ (mm)  |
|---------------|--------------|-------|-------|----------|-------|-------|--------------|-------------|----------|-------|-------|--------------|-------------|
|               | X(mm)        | Y(mm) | Z(mm) | X(mm)    | Y(mm) | Z(mm) |              |             | X(mm)    | Y(mm) | Z(mm) |              |             |
| 1             | 5            | -70   | -500  | 4        | -71   | -500  | <b>1.41</b>  | <b>1.41</b> | 4        | -72   | -509  | 9.27         | 2.24        |
|               | -39          | -70   | -498  | -38      | -71   | -487  | 11.09        | 1.41        | -39      | -71   | -495  | <b>3.16</b>  | <b>1.00</b> |
|               | -33          | -90   | -502  | -31      | -87   | -475  | 27.24        | 3.61        | -32      | -90   | -495  | <b>7.07</b>  | <b>1.00</b> |
| 2             | 108          | -70   | -500  | 106      | -69   | -490  | <b>10.25</b> | 2.24        | 107      | -72   | -512  | 12.21        | <b>2.24</b> |
|               | 64           | -70   | -507  | 62       | -70   | -491  | 16.12        | 2.00        | 64       | -70   | -495  | <b>12.00</b> | <b>0.00</b> |
|               | 68           | -70   | -520  | 68       | -69   | -495  | 25.02        | <b>1.00</b> | 69       | -68   | -512  | <b>8.31</b>  | 2.24        |
| 3             | 211          | -120  | -500  | 207      | -119  | -490  | 10.82        | 4.12        | 209      | -119  | -495  | <b>5.48</b>  | <b>2.24</b> |
|               | 166          | -120  | -516  | 164      | -122  | -515  | 3.00         | 2.83        | 167      | -122  | -516  | <b>2.24</b>  | <b>2.24</b> |
|               | 170          | -120  | -538  | 163      | -121  | -526  | 13.93        | 7.07        | 166      | -119  | -532  | <b>7.28</b>  | <b>4.12</b> |
| 4             | 114          | -120  | -500  | 110      | -117  | -510  | 11.18        | 5.00        | 113      | -118  | -492  | <b>8.31</b>  | <b>2.24</b> |
|               | 70           | -120  | -508  | 66       | -107  | -466  | 44.11        | 13.60       | 71       | -118  | -494  | <b>14.07</b> | <b>2.24</b> |
|               | 74           | -120  | -521  | 73       | -114  | -481  | 40.46        | <b>6.08</b> | 75       | -109  | -485  | <b>37.62</b> | 11.05       |
| 5             | 5            | -67   | -500  | 4        | -66   | -496  | <b>4.24</b>  | <b>1.41</b> | 4        | -68   | -510  | 10.10        | 1.41        |
|               | -39          | -67   | -498  | -39      | -66   | -491  | 7.07         | <b>1.00</b> | -38      | -68   | -496  | <b>2.45</b>  | 1.41        |
|               | -33          | -67   | -502  | -30      | -69   | -476  | 26.25        | 3.61        | -31      | -64   | -491  | <b>11.45</b> | 3.61        |
| 6             | 108          | -67   | -500  | 106      | -65   | -491  | 9.43         | 2.83        | 108      | -67   | -499  | <b>1.00</b>  | <b>0.00</b> |
|               | 64           | -67   | -507  | 62       | -66   | -490  | 17.15        | 2.24        | 64       | -66   | -495  | <b>12.04</b> | <b>1.00</b> |
|               | 68           | -67   | -520  | 66       | -63   | -477  | 43.16        | 4.47        | 70       | -66   | -505  | <b>15.07</b> | <b>2.24</b> |
| 7             | 211          | -67   | -500  | 207      | -66   | -490  | 10.82        | 4.12        | 209      | -64   | -495  | <b>5.92</b>  | <b>3.61</b> |
|               | 166          | -67   | -516  | 164      | -68   | -509  | 7.35         | 2.24        | 165      | -65   | -516  | <b>2.24</b>  | 2.24        |
|               | 170          | -67   | -538  | 163      | -70   | -526  | 14.04        | 7.62        | 169      | -69   | -537  | <b>2.45</b>  | <b>2.24</b> |
| 8             | 114          | -67   | -500  | 112      | -67   | -507  | 7.28         | 2.00        | 113      | -66   | -493  | <b>7.14</b>  | <b>1.41</b> |
|               | 20           | -67   | -504  | 20       | -67   | -492  | <b>12.00</b> | <b>0.00</b> | 21       | -66   | -491  | 13.19        | 1.41        |
|               | 74           | -67   | -521  | 73       | -62   | -469  | 52.24        | 5.10        | 74       | -63   | -478  | <b>43.05</b> | <b>4.00</b> |
| 9             | 5            | -14   | -500  | 2        | -17   | -516  | 16.55        | 4.24        | 5        | -13   | -498  | <b>2.24</b>  | <b>1.00</b> |
|               | -39          | -14   | -498  | -39      | -14   | -494  | <b>4.00</b>  | <b>0.00</b> | -38      | -14   | -491  | 7.07         | 1.00        |
|               | -33          | -14   | -502  | -35      | -15   | -500  | <b>3.00</b>  | <b>2.24</b> | -29      | -13   | -478  | 24.35        | 4.12        |
| 10            | 108          | -14   | -500  | 106      | -15   | -501  | <b>2.45</b>  | 2.24        | 106      | -14   | -509  | 9.22         | <b>2.00</b> |
|               | 64           | -14   | -507  | 62       | -13   | -483  | 24.10        | 2.24        | 64       | -13   | -492  | <b>15.03</b> | <b>1.00</b> |
|               | 68           | -14   | -520  | 66       | -14   | -477  | 43.05        | 2.00        | 69       | -14   | -509  | <b>11.05</b> | <b>1.00</b> |
| 11            | 211          | -14   | -500  | 209      | -14   | -501  | <b>2.24</b>  | <b>2.00</b> | 209      | -14   | -495  | 5.39         | 2.00        |
|               | 166          | -14   | -516  | 164      | -15   | -515  | <b>2.45</b>  | <b>2.24</b> | 163      | -14   | -514  | 3.61         | 3.00        |
|               | 22           | -14   | -512  | 22       | -14   | -492  | 20.00        | <b>0.00</b> | 18       | -12   | -522  | <b>10.95</b> | 4.47        |
| 12            | 64           | -14   | -500  | 63       | -13   | -500  | <b>1.41</b>  | 1.41        | 64       | -13   | -492  | 8.06         | <b>1.00</b> |
|               | 70           | -14   | -508  | 67       | -14   | -477  | 31.14        | 3.00        | 70       | -14   | -504  | <b>4.00</b>  | <b>0.00</b> |
|               | -24          | -14   | -503  | -22      | -13   | -491  | <b>12.21</b> | <b>2.24</b> | -19      | -13   | -475  | 28.46        | 5.10        |
| <b>Mean</b>   |              |       |       |          |       |       | <b>16.34</b> | <b>3.08</b> |          |       |       | <b>10.63</b> | <b>2.31</b> |
| <b>Median</b> |              |       |       |          |       |       | <b>11.59</b> | <b>2.24</b> |          |       |       | <b>8.31</b>  | <b>2.24</b> |

Table A.3

Orientation (quaternion) comparison between the proposed method and the AprilTag-based baseline.

| Scenario | Ground truth(wxyz)      | Proposed(wxyz)             | AprilTag(wxyz)             | $e_{\theta}$ proposed (°) | $e_{\theta}$ AprilTag (°) |
|----------|-------------------------|----------------------------|----------------------------|---------------------------|---------------------------|
| 1        | 1.000 0.000 0.000 0.000 | 0.997 0.057 0.044 -0.009   | 0.997 0.056 0.046 -0.010   | <b>8.33</b>               | <b>8.39</b>               |
|          | 0.999 0.000 0.044 0.000 | 0.999 0.018 0.029 -0.005   | 0.999 0.007 0.032 -0.004   | 2.75                      | <b>1.66</b>               |
|          | 0.996 0.000 0.087 0.000 | 0.999 0.020 0.026 -0.005   | 0.997 0.026 0.071 -0.005   | 7.39                      | <b>3.55</b>               |
| 2        | 1.000 0.000 0.000 0.000 | 0.998 0.061 -0.002 0.000   | 0.992 -0.111 0.055 -0.003  | <b>7.01</b>               | 15.97                     |
|          | 0.999 0.000 0.044 0.000 | 1.000 0.006 0.003 -0.000   | 1.000 -0.003 0.015 -0.000  | <b>4.56</b>               | 3.37                      |
|          | 0.996 0.000 0.087 0.000 | 1.000 -0.006 -0.003 0.000  | 0.996 0.049 0.081 0.003    | <b>10.37</b>              | 5.57                      |
| 3        | 1.000 0.000 0.000 0.000 | 1.000 0.021 -0.016 0.004   | 1.000 -0.005 0.007 -0.002  | 2.92                      | <b>1.03</b>               |
|          | 0.999 0.000 0.044 0.000 | 0.999 -0.036 0.027 -0.005  | 1.000 -0.001 0.018 -0.001  | 5.38                      | <b>3.03</b>               |
|          | 0.996 0.000 0.087 0.000 | 0.990 -0.074 0.117 -0.018  | 0.993 0.009 0.121 -0.009   | <b>9.76</b>               | 10.39                     |
| 4        | 1.000 0.000 0.000 0.000 | 0.991 -0.103 0.082 -0.010  | 1.000 0.000 0.002 -0.001   | 17.19                     | <b>0.26</b>               |
|          | 0.999 0.000 0.044 0.000 | 0.992 -0.085 -0.097 0.014  | 1.000 -0.002 0.012 0.000   | 17.55                     | <b>3.74</b>               |
|          | 0.996 0.000 0.087 0.000 | 0.996 -0.076 -0.047 -0.002 | 0.971 -0.226 -0.074 0.031  | <b>15.43</b>              | 28.88                     |
| 5        | 1.000 0.000 0.000 0.000 | 1.000 0.006 -0.006 0.001   | 1.000 0.000 -0.000 -0.001  | 0.99                      | <b>0.11</b>               |
|          | 0.999 0.000 0.044 0.000 | 1.000 0.003 -0.000 0.000   | 0.995 -0.075 0.059 -0.005  | <b>4.88</b>               | 9.66                      |
|          | 0.996 0.000 0.087 0.000 | 0.992 -0.066 0.102 -0.016  | 0.998 -0.011 0.063 -0.006  | <b>9.10</b>               | 7.12                      |
| 6        | 1.000 0.000 0.000 0.000 | 1.000 0.013 0.016 -0.002   | 0.998 0.033 0.053 -0.006   | <b>2.98</b>               | 7.24                      |
|          | 0.999 0.000 0.044 0.000 | 1.000 0.014 0.025 -0.004   | 0.999 0.017 0.044 -0.005   | <b>3.18</b>               | 2.78                      |
|          | 0.996 0.000 0.087 0.000 | 0.999 0.019 0.029 -0.012   | 0.997 -0.056 0.057 0.013   | <b>6.84</b>               | 10.35                     |
| 7        | 1.000 0.000 0.000 0.000 | 1.000 0.018 -0.001 0.004   | 1.000 -0.003 0.006 -0.000  | 2.12                      | <b>0.77</b>               |
|          | 0.999 0.000 0.044 0.000 | 1.000 -0.023 -0.021 0.002  | 1.000 -0.004 0.016 -0.000  | 8.50                      | <b>3.22</b>               |
|          | 0.996 0.000 0.087 0.000 | 0.996 -0.045 -0.072 0.006  | 0.998 0.047 0.044 0.001    | 21.31                     | <b>8.71</b>               |
| 8        | 1.000 0.000 0.000 0.000 | 1.000 0.012 -0.016 0.002   | 0.998 0.055 0.011 0.011    | <b>2.40</b>               | 7.25                      |
|          | 0.999 0.000 0.044 0.000 | 0.999 -0.026 0.038 -0.004  | 0.997 0.027 0.071 0.004    | <b>1.37</b>               | 3.24                      |
|          | 0.996 0.000 0.087 0.000 | 0.992 -0.043 0.118 -0.011  | 0.995 0.005 0.097 -0.010   | 4.34                      | <b>2.40</b>               |
| 9        | 1.000 0.000 0.000 0.000 | 0.998 -0.054 0.041 -0.003  | 0.999 0.037 0.005 -0.000   | 8.29                      | <b>4.30</b>               |
|          | 0.999 0.000 0.044 0.000 | 1.000 0.023 0.021 -0.003   | 1.000 0.003 0.005 0.001    | 2.82                      | <b>4.49</b>               |
|          | 0.996 0.000 0.087 0.000 | 0.994 -0.051 -0.092 0.008  | 0.995 -0.039 -0.092 0.011  | 20.63                     | <b>19.26</b>              |
| 10       | 1.000 0.000 0.000 0.000 | 1.000 0.007 -0.011 0.006   | 1.000 0.015 -0.021 0.002   | <b>1.64</b>               | 3.12                      |
|          | 0.999 0.000 0.044 0.000 | 0.999 0.037 -0.007 0.001   | 0.996 -0.036 0.085 -0.006  | <b>4.91</b>               | 5.23                      |
|          | 0.996 0.000 0.087 0.000 | 0.998 -0.030 0.063 -0.006  | 0.969 -0.124 -0.214 0.009  | <b>2.83</b>               | 33.19                     |
| 11       | 1.000 0.000 0.000 0.000 | 0.994 0.068 0.080 -0.008   | 1.000 -0.017 0.015 0.002   | 11.21                     | <b>2.61</b>               |
|          | 0.999 0.000 0.044 0.000 | 0.999 0.004 0.034 -0.001   | 1.000 -0.014 0.028 0.001   | <b>1.34</b>               | 2.72                      |
|          | 0.996 0.000 0.087 0.000 | 0.998 0.013 0.065 -0.003   | 1.000 -0.011 0.027 0.002   | <b>2.73</b>               | 7.79                      |
| 12       | 1.000 0.000 0.000 0.000 | 0.998 0.055 -0.003 0.000   | 0.997 0.007 0.082 -0.002   | <b>6.31</b>               | 9.42                      |
|          | 0.999 0.000 0.044 0.000 | 0.999 -0.007 -0.039 0.001  | 1.000 -0.030 0.003 0.001   | 9.55                      | <b>4.85</b>               |
|          | 0.996 0.000 0.087 0.000 | 0.997 -0.010 -0.072 0.001  | 0.997 -0.016 0.076 -0.001  | 18.19                     | <b>1.29</b>               |
| 13       | 1.000 0.000 0.000 0.000 | 1.000 0.004 -0.024 0.001   | 1.000 -0.005 0.010 -0.002  | 2.75                      | <b>1.31</b>               |
|          | 0.999 0.000 0.044 0.000 | 1.000 -0.004 0.023 -0.001  | 0.996 -0.010 0.091 -0.003  | <b>2.48</b>               | 5.74                      |
|          | 0.996 0.000 0.087 0.000 | 1.000 0.005 0.020 -0.001   | 0.982 0.024 0.187 0.002    | <b>7.68</b>               | 11.56                     |
| 14       | 1.000 0.000 0.000 0.000 | 1.000 -0.003 -0.010 0.005  | 1.000 0.006 -0.003 0.000   | 1.33                      | <b>0.77</b>               |
|          | 0.999 0.000 0.044 0.000 | 0.997 -0.010 -0.073 0.001  | 0.998 0.001 0.056 -0.001   | 13.47                     | <b>1.39</b>               |
|          | 0.996 0.000 0.087 0.000 | 0.999 0.009 0.053 0.002    | 1.000 -0.002 0.018 -0.001  | 3.89                      | <b>7.93</b>               |
| 15       | 1.000 0.000 0.000 0.000 | 1.000 -0.001 0.006 -0.000  | 1.000 -0.012 -0.014 -0.001 | <b>0.70</b>               | 2.12                      |
|          | 0.999 0.000 0.044 0.000 | 1.000 -0.002 0.009 -0.000  | 0.998 0.001 0.056 -0.001   | 4.02                      | <b>1.39</b>               |
|          | 0.996 0.000 0.087 0.000 | 0.996 -0.011 -0.088 0.002  | 1.000 -0.002 0.018 -0.001  | 20.12                     | <b>7.93</b>               |
| Mean     |                         |                            |                            | <b>7.19</b>               | <b>6.38</b>               |
| Median   |                         |                            |                            | <b>4.91</b>               | <b>4.30</b>               |

## References

- Abioye, S. O., Oyedele, L. O., Akanbi, L., Ajayi, A., Davila Delgado, J. M., Bilal, M., Akinade, O. O., & Ahmed, A. (2021). Artificial intelligence in the construction industry: A review of present status, opportunities and future challenges. *Journal of Building Engineering*, 44, 103299. <https://doi.org/10.1016/j.jobe.2021.103299>
- Behnam, M. T., BuHamdan, S., & Alwisay, A. (2023). Robotics in assembly-based industrialized construction: A narrative review and a look forward. *International Journal of Intelligent Robotics and Applications*, 7(3), 556–574.
- Boldo, M., De Marchi, M., Martini, E., Aldegheri, S., Quaglia, D., Fummi, F., & Bombieri, N. (2024). Real-time multi-camera 3D human pose estimation at the edge for industrial applications. *Expert Systems with Applications*, 252, 124089.
- Cărunta, C., Cărunta, A., & Popa, C.-A. (2025). Heavy and lightweight deep learning models for semantic segmentation: A survey. *IEEE Access*, 13, 17745–17765. <https://doi.org/10.1109/ACCESS.2025.3529812>
- Chen, B., Niu, T., Yu, W., Zhang, R., Wang, Z., & Li, B. (2023). A-Net: An a-shape lightweight neural network for real-time surface defect segmentation. *IEEE Transactions on Instrumentation and Measurement*, 73, 1–14.
- Chen, L., Wen, J., Gao, Y., Yang, J., Zhang, C., An, S., & Wu, P. (2026). Edge-guided semi-supervised 6D pose estimation with cross-domain alignment for robotic grasping. *Expert Systems with Applications*, 298, 129880. <https://doi.org/10.1016/j.eswa.2025.129880>
- Chen, L.-C., Papandreou, G., Kokkinos, I., Murphy, K., & Yuille, A. L. (2015). Semantic image segmentation with deep convolutional nets and fully connected CRFs. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Chen, L.-C., Zhu, Y., Papandreou, G., Schroff, F., & Adam, H. (2018a). Encoder-decoder with atrous separable convolution for semantic image segmentation. In *Proceedings of the European conference on computer vision (ECCV)* (pp. 801–818).
- Chen, Q., de Soto, B. G., & Adey, B. T. (2018b). Construction automation: Research areas, industry concerns and suggestions for advancement. *Automation in construction*, 94, 22–38.
- Davila Delgado, J. M., Oyedele, L., Ajayi, A., Akanbi, L., Akinade, O., Bilal, M., & Owolabi, H. (2019). Robotics and automated systems in construction: Understanding industry-specific challenges for adoption. *Journal of Building Engineering*, 26, 100868. <https://doi.org/10.1016/j.jobe.2019.100868>
- Deng, Z., Luo, H., Gao, X., & Liu, P. (2025). Fusion-based localization system integrating UWB, IMU, and vision. *Applied Sciences*, 15(12), 6501. <https://doi.org/10.3390/app15126501>
- Feng, C., Yu, J., Li, J., Wang, K., Wang, B., & Wu, Y. (2025). Evaluation of LiDAR SLAM algorithms for construction robots in large public construction sites. *Low-Carbon Materials and Green Construction*, 3, 1, 28. <https://doi.org/10.1007/s44242-025-00092-8>
- Formisano, A., & Naqash, M. T. (2026). Innovative reinforced aluminium alloy curtain walls: Numerical analysis and experimental validation. *Journal of Building Engineering*, 117, 114554. <https://doi.org/10.1016/j.jobe.2025.114554>
- Guo, M.-H., Lu, C.-Z., Hou, Q., Liu, Z., Cheng, M.-M., & Hu, S.-M. (2022). SEGNext: Rethinking convolutional attention design for semantic segmentation. *Advances in Neural Information Processing Systems*, 35, 1140–1156.
- Halder, S., & Afsari, K. (2023). Robots in inspection and monitoring of buildings and infrastructure: A systematic review. *Applied Sciences*, 13(4). <https://doi.org/10.3390/app13042304>
- Hassen, A. Y., Arashpour, M., & Abdi, E. (2025). Lightweight segmentation model for automated facade installation in high-rise buildings. *Advanced Engineering Informatics*, 65, 103374.
- Hong, Y., Pan, H., Sun, W., & Jia, Y. (2021). Deep dual-resolution networks for real-time and accurate semantic segmentation of road scenes. *arXiv preprint arXiv:2101.06085*.
- Huang, H., Ho, G. T. S., Tang, V., Chow, E. W. H., Ip, W. H., & Yung, K. L. (2026). A lightweight fully convolutional neural network for efficient and accurate tunnel crack detection. *Journal of Advanced Research*, 86, 511–525. <https://doi.org/10.1016/j.jare.2025.11.052>
- Iturralde, K., Feucht, M., Illner, D., Hu, R., Pan, W., Linner, T., Bock, T., Eskudero, I., Rodriguez, M., Gorrotxategi, J., Izard, J. B., Astudillo, J., Cavalcanti Santos, J., Gouttefarde, M., Fabritius, M., Martin, C., Henning, T., Nornes, S. M., Jacobsen, Y., Pracucci, A., Cañada, J., Jimenez-Vicaria, J. D., Alonso, R., & Elia, L. (2022). Cable-driven parallel robot for curtain wall module installation. *Automation in Construction*, 138, 104235. <https://doi.org/10.1016/j.autcon.2022.104235>

- Johns, B., Abdi, E., & Arashpour, M. (2023a). Crane payload localisation for curtain wall installation: A markerless computer vision approach. *Measurement*, 221, 113459. <https://doi.org/10.1016/j.measurement.2023.113459>
- Johns, B., Abdi, E., & Arashpour, M. (2023b). Glass curtain wall installation dataset. <https://github.com/Brandon-Johns/glass-curtain-wall-installation-dataset>, <https://doi.org/10.26180/23538198>
- Johns, B., Arashpour, M., & Abdi, E. (2020). Curtain wall installation for high-rise buildings: Critical review of current automation solutions and opportunities. In *Proceedings of the 37th international symposium on automation and robotics in construction, ISARC 2020* (pp. 393–400). International Association for Automation and Robotics in Construction (IAARC). <https://isarc2020.org>
- Kalaitzakis, M., Cain, B., Carroll, S., Ambrosi, A., Whitehead, C., & Vitzilaos, N. (2021). Fiducial markers for pose estimation: Overview, applications and experimental comparison of the ARTag, apriltag, aruco and STag markers. *Journal of Intelligent & Robotic Systems*, 101(4), 71. <https://doi.org/10.1007/s10846-020-01307-9>
- Kayhani, N., Zhao, W., McCabe, B., & Schoellig, A. P. (2022). Tag-based visual-inertial localization of unmanned aerial vehicles in indoor construction environments using an on-manifold extended kalman filter. *Automation in Construction*, 135, 104112. <https://doi.org/10.1016/j.autcon.2021.104112>
- Kim, J., Chi, S., & Kim, J. (2023). 3D pose estimation and localization of construction equipment from single camera images by virtual model integration. *Advanced Engineering Informatics*, 57, 102092. <https://doi.org/10.1016/j.aei.2023.102092>
- Krogius, M., Haggenmiller, A., & Olson, E. (2019). Flexible layouts for fiducial tags. In *Proceedings of the IEEE/RSJ international conference on intelligent robots and systems (IROS)*.
- Lee, Y.-C. (2024). Lsmcl: Long-term static mapping and cloning localization for autonomous robot navigation using 3d lidar in dynamic environments. *Expert Systems with Applications*, 241, 122688.
- Lee, Y.-S., Kim, S.-H., Gil, M.-S., Lee, S.-H., Kang, M.-S., Jang, S.-H., Yu, B.-H., Ryu, B.-G., Hong, D., & Han, C.-S. (2018). The study on the integrated control system for curtain wall building façade cleaning robot. *Automation in Construction*, 94, 39–46. <https://doi.org/10.1016/j.autcon.2017.12.030>
- Liang, Z., Hu, Y., Fang, T., Wang, Y., He, X., & Tian, T. (2026). Multi-source feature contrast learning network for weakly-supervised semantic segmentation. *Expert Systems with Applications*, 299, 130131. <https://doi.org/10.1016/j.eswa.2025.130131>
- Liu, X., Cheng, Y., Wang, W., Huang, T., & Feng, W. (2025). Dynamic parameter identification of a curtain wall installation robotic arm. *arXiv preprint arXiv:2507.17136*.
- Liu, Z., Mao, H., Wu, C.-Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 11976–11986).
- Luo, L., Yan, B., Wu, Y., Ouyang, K., & Li, S. (2025). Full-field deformation measurement of glass curtain walls and structural-adhesive damage detection using 3D-DIC method. *Structures*, 78, 109277. <https://doi.org/10.1016/j.istruc.2025.109277>
- MMSEG (2020). MMSegmentation: Openmmlab semantic segmentation toolbox and benchmark. <https://github.com/open-mmlab/mms Segmentation>.
- Oesterreich, T. D., & Teuteberg, F. (2016). Understanding the implications of digitisation and automation in the context of industry 4.0: A triangulation approach and elements of a research agenda for the construction industry. *Computers in Industry*, 83, 121–139. <https://doi.org/10.1016/j.compind.2016.09.006>
- Oršić, M., & Šegvić, S. (2021). Efficient semantic segmentation with pyramidal fusion. *Pattern Recognition*, 110, 107611.
- Ouda, E., & Haggag, M. (2024). Automation in modular construction manufacturing: A comparative analysis of assembly processes. *Sustainability*, 16(21). <https://www.mdpi.com/2071-1050/16/21/9238>
- Paszke, A., Chaurasia, A., Kim, S., & Culurciello, E. (2016). ENet: A deep neural network architecture for real-time semantic segmentation. *arXiv preprint arXiv:1606.02147*.
- Peng, J., Liu, Y., Tang, S., Hao, Y., Chu, L., Chen, G., Wu, Z., Chen, Z., Yu, Z., Du, Y., Dang, Q., Lai, B., Liu, Q., Hu, X., Yu, D., & Ma, Y. (2022). PP-LiteSeg: A superior real-time semantic segmentation model. <https://doi.org/10.48550/arXiv.2204.02681>
- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In *International conference on medical image computing and computer-assisted intervention* (pp. 234–241). Springer.
- Shi, M., Shen, J., Yi, Q., Weng, J., Huang, Z., Luo, A., & Zhou, Y. (2022). LMFFNet: A well-balanced lightweight network for fast and accurate semantic segmentation. *IEEE Transactions on Neural Networks and Learning Systems*, 34(6), 3205–3219.
- Stührenberg, J., & Smarsly, K. (2025). LIO-BIM: Coupling lidar inertial odometry with building information modeling for robot localization and mapping. *Advanced Engineering Informatics*, 66, 103477. <https://doi.org/10.1016/j.aei.2025.103477>
- Taghavi, M., Heredia, H., Iturralde, K., Halvorsen, H., & Bock, T. (2018). Development of a modular end effector for the installation of curtain walls with cable-robots. *Journal of Facade Design and Engineering*, 6(2), 001–008. <https://jfd.eu/index.php/jfde/article/view/161>, <https://doi.org/10.7480/jfde.2018.2.2067>
- Tang, J., Luo, H., Chen, W., Wong, P. K.-Y., & Cheng, J. C. P. (2022). IMU-based full-body pose estimation for construction machines using kinematics modeling. *Automation in Construction*, 138, 104217. <https://doi.org/10.1016/j.autcon.2022.104217>
- Todorov, E., Erez, T., & Tassa, Y. (2012). Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ International conference on intelligent robots and systems* (pp. 5026–5033). IEEE. <https://doi.org/10.1109/IROS.2012.6386109>
- Wang, H., Chen, Y., Liu, Q., Gao, M., Lu, X., Zhang, Z., & Dong, Q. (2026). Multi-view consistent feature learning for open-set semantic image segmentation. *Expert Systems with Applications*, 305, 130791. <https://doi.org/10.1016/j.eswa.2025.130791>
- Wang, J., Sun, K., Cheng, T., Jiang, B., Deng, C., Zhao, Y., Liu, D., Mu, Y., Tan, M., Wang, X. et al. (2020). Deep high-resolution representation learning for visual recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(10), 3349–3364.
- Wang, L., Zhang, M., Li, H., Hu, Y., Ma, J., Umer, W., & Fang, X. (2026). Modeling trust in human-robot collaborative construction: An improved cloud bayesian network. *Expert Systems with Applications*, 298, 129928.
- Wu, D., Xu, X., Li, R., Peng, X., Gong, X., Lee, C.-H., Pan, P., & Jiang, S. (2024). Curtain wall frame segmentation using a dual-flow aggregation network: Application to robot pose estimation. *Automation in Construction*, 168, 105816. <https://doi.org/10.1016/j.autcon.2024.105816>
- Wu, H., Li, H., Fang, X., & Luo, X. (2022). A survey on teaching workplace skills to construction robots. *Expert Systems with Applications*, 205, 117658.
- Xu, J., Xiong, Z., & Bhattacharyya, S. P. (2023). PIDNet: A real-time semantic segmentation network inspired by PID controllers. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 19529–19539).
- Yang, L., & Cai, H. (2024). Enhanced visual SLAM for construction robots by efficient integration of dynamic object segmentation and scene semantics. *Advanced Engineering Informatics*, 59, 102313. <https://doi.org/10.1016/j.aei.2023.102313>
- Yu, C., Gao, C., Wang, J., Yu, G., Shen, C., & Sang, N. (2021). BiseNet v2: Bilateral network with guided aggregation for real-time semantic segmentation. *International Journal of Computer Vision*, 129(11), 3051–3068.
- Yu, C., Wang, J., Peng, C., Gao, C., Yu, G., & Sang, N. (2018). BiseNet: Bilateral segmentation network for real-time semantic segmentation. In *Proceedings of the European conference on computer vision (ECCV)* (pp. 325–341).
- Yu, S. N., Lee, S. Y., Han, C. S., Lee, K. Y., & Lee, S. H. (2007). Development of the curtain wall installation robot: Performance and efficiency tests at a construction site. *Autonomous Robots*, 22(3), 281–291.
- Zhang, M., Xie, T., Fang, Y., Zang, X., Zhao, J., & Zhang, X. (2026). EFT6D: An efficient fusion transformer network for 6D object pose estimation. *Expert Systems with Applications*, 296, 129035. <https://doi.org/10.1016/j.eswa.2025.129035>
- Zheng, S., Lu, J., Zhao, H., Zhu, X., Luo, Z., Wang, Y., Fu, Y., Feng, J., Xiang, T., Torr, P. H. S. et al. (2021). Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 6881–6890).